REPORT OF ACTIVITIES

2012

Institut de Ciències del Cosmos

Universitat de Barcelona

B:KC Barcelona Knowledge Campus
Campus d'Excel·lència Internacional

Hopefully, this year 2012 will be the peak of the crisis that is severely affecting the funding of research projects by the Spanish Ministry of Economics and Competitiveness (MINECO) as well as the endowment of the associated fellowships and contracts. Similarly, the new rules that govern the implementation of the Ramon y Cajal programme are so stringent that Spanish universities and CSIC will hardly be able to offer new positions. This is a big problem that certainly threatens to roll back the Spanish science to more than 10 years ago and thus makes Spain waste a unique opportunity of reaching the R&D level it deserves.

The news from the Universitat de Barcelona (UB) is equally bad. Financial straits have forced the UB to make a sever cut, of about 30 %, in the budget devoted to the Contract Programmes with its Instituts Propis de Recerca (University Institutes). And this is not all. Researchers in the last year of a Ramon y Cajal (RyC) contract at the UB will have next year the opportunity to apply to a Serra Hunter position, but it is not clear when this would take place. Such a wait is a dramatic unexpected situation for all these top-quality researchers. The future of the RyC contracts at the UB is something of a great concern as this is the only way for young talented researchers to access the Spanish science system, in particular our Institute.

Fortunately, there is also good news. Firstly, I am once more glad to mention the success of the ICC-UB in its participation in the 2011 call of the Severo Ochoa award to the Spanish research centres of excellence. The ICC-UB has again been finalist. Of course, we would rather prefer to have been directly awarded, but we were closer than ever to be so. We were indeed among the very few centres reaching the necessary score of 95 points over one hundred. Only the reduction this year by a factor two of the total number of awarded centres prevented us for obtaining the prize. This result is particularly remarkable if we take into consideration that, in this second call, all the candidate research centres were more experienced and had more information about it, so the quality of the applications was notably higher than the first year.

Secondly, other facts that occurred during 2012 make us be particularly proud of the ICC-UB. On the one hand, Licia Verde, as a member of the team led by Charles Benett, obtained the Gruber Prize of Cosmology for their analysis, in 2003, of the cosmic microwave background (CMB) anisotropies measured by the WMAP satellite. This work has meant the beginning of the so-called "cosmology precision era". I want to remark that this is the second time that an ICC-UB member is awarded such a prestigious prize, considered to be an anteroom of the Nobel Prize. L. Verde plaid a crucial role in that work, which has already got more than 6,000 cites. In fact, L. Verde, with more than 20,000 cites (SAO/NASA Astrophysics Data System) is the most cited young female astrophysicist today in the world. On the other hand, David Mateos has been awarded with one ERC Starting Grant for the study of quantum chromo-dynamics using gauge/string duality. This ERC Starting Grant is to be added to that previously obtained by L. Verde.

All these very positive results show that, despite of the economic difficulties we are going through, the ICC-UB continues to be in the good track.

Eduard Salvador-Solé
Director

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Vidaña, Isaac
Vieiro, Arturo

RESEARCH ACTIVITY

Cosmology and Large Scale Structure

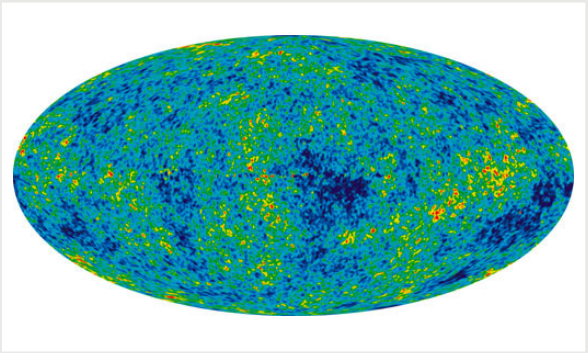
One of the main interests at the ICCUB is the study of the connection between cosmological observations and the physics behind the standard cosmological model, hoping to shed some light on the "open questions" in cosmology.

Cosmology confronted to astronomical observations is mostly carried out by researchers of the Astrophysics and Space Science Section. Their research ranges from the inflationary model, the Cosmic Microwave Background (CMB) and the epoch of reionization, to the formation and evolution of galaxies and the distribution of gas in space, including statistical applications and data analysis. Research is also being carried out into the nature of dark matter and the primordial fluctuations that gave rise to galaxies and larger structures in the universe. The observational tools available for these studies are gravitational lensing by galaxies, clusters of galaxies and large-scale structure, microlensing of stars or quasars by any kind of compact object, the spatial distribution of galaxies and of matter in intergalactic space that is measured from absorption signatures in spectra of background sources, and the structure of dark matter halos studied through the dynamics of galaxies in clusters.

The ICCUB is also involved in several cosmological projects of ground-based and space surveys, such as SSDS-III, the EUCLID science working group, the CORE science working group and the LSST large scale structure study group.

Members

Ariño, Andreu; Arnau, Eduard; Canal, Ramon; de Putter, Roland; Hoyle, Ben; Jiménez, Raul; Juan, Enric; Labay, J. Javier; Manrique, Alberto; Miralda-Escudé, Jordi; Noreña, Jorge; Pérez, Ignasi; Prieto, Joaquin; Robaina, Aday; Ruiz-Lapuente, M. Pilar; Sala, Ferran; Salvador-Solé, Eduard; Verde, Licia; Viñas, Jordi; Wagner, Christian; Zumalacarregui, Miguel.



Map of the CMB created from data gathered by the Wilkinson Microwave Anisotropy Probe (WMAP).

The image reveals 13.77 billion year old temperature fluctuations (shown as color differences) that correspond to the seeds that grew to become galaxies. **Credit:** NASA/WMAP Science Team.

Lines of Research

Large scale structure of galaxies and the intergalactic medium.

Microwave background radiation anisotropies.

Baryonic acoustic oscillations.

Supernova cosmology.

Dark matter and dark energy.

Lyman-alpha emission from galaxies at high redshifts.

Reionization of the intergalactic medium.

Experimental Particle Physics

The ICCUB's experimental particle physicists are specialized in the study of flavor physics. Specifically in measuring CP violation effects and rare decays of particles containing b or c quaks. Their long track record in this field go back into the design, construction and exploitation in the Hera-B experiment at the Desy lab, in Hamburg and the participation in the BaBar experiment at the SLAC National lab, at Stanford, CA.

Currently the group is fully involved in the LHCb experiment data analysis and on its upgrade project. The LHCb detector, one of the four detectors of the Large Hadron Collider in CERN (Geneva), is designed to study this asymmetry through the b and anti-b particle pairs produced in proton collisions. The ICCUB, aside from its participation at a scientific level, undertook the design, production and installation of the electronics of the SPD (Scintillator Pad Detector) part of the calorimeter. In addition, the ICCUB participated in the development of the Data-GRID computer network and the DIRAC software, which are essential in the data analysis and simulation processes, not only in LHCb, but also in all HEP experiment nowadays.

An updated LHCb detector is currently being designed and scheduled for 2018 to start operation. The ICCUB participates in the design of the readout (RO) electronics of both the calorimeter and the new central tracker (to be based on Scintillating Fibers).

Members

Potterat, Cédric; Rives Molina, Vicente; Garrido, Lluís; Graciani Díaz, Ricardo; Graugés, Eugeni; Ruiz, Hugo; Casajús, Adrià; Comerma-Montells, Albert; Lazovski, Nikola; Picatoste Olloqui, Eduard; Sanuy, Andreu; Trenado, Juan; Gascón Fora, David.

The LHCb detector

The LHCb is one of the 4 detectors of the Large Hadron Collider (LHC). It is a single arm forward spectrometer designed to study CP violation effects and rare decays of hadrons containing b or c quarks. **Credit:** CERN.

Lines of Research

Physics of beauty and charm mesons.

Charge-Parity (CP) symmetry violation.

Search for deviations from the Standard Model in rare B (and charm) meson decays.

Quarkonium.

Development of distributed calculation methods using grid and cloud computing.

Design of Geiger mode avalanche photodiodes for tracking detectors of future accelerators.

Simulation and study of the radiation hardness of avalanche photodetectors.

Design, contruction and operation of instrumentation for high energy, astrophysics and medical imaging experiments.

Galactic Astronomy

Research in galactic astronomy at the ICCUB is focused on the preparation of the scientific exploitation of the Gaia mission. The ICCUB's researchers are both developing detailed galaxy models and participating in huge on-ground spectroscopic survey complementary to Gaia.

The stellar clusters, excellent tracers of the chemo-dynamical evolution of the galactic disk, are being deeply analyzed. The connection between the stellar formation processes and the spiral arms and bar evolution is being addressed. New methods for accurate derivation of critical parameters such us ages and metallicities are being developed. In the Gaia era it is fundamental to establish a stellar luminosity calibration. Robust statistical techniques are being developed to achieve unbiased determination of stellar distances.

The ICCUB participates in an EC FP7 funded initiative named GREAT-ITN (2011-2015) aiming to form the next generation of experts in this field.

The ICCUB is also leading the Spanish Network for Gaia Science Exploitation (REG) with more than 140 Spanish researchers from 24 institutions.

Extragalactic Astronomy

The ICCUB's interest in galactic astrophysics extends beyond the Milky Way and is concerned, too, with the formation of the first galaxies, which were formed from pristine matter. They comprised population III stars, which reionised the intergalactic medium, polluted it with metals and left behind the seeds of super massive black holes. These are processes currently being modeled at the ICCUB.

Detailed analytical models and huge numerical simulations are being developed which make use of the most powerful computational tools presently available. The resulting predictions are confronted with the latest, progressively complete, observations drawn from huge wide angle (all-sky) nearby galaxy surveys (e.g. SDSS, 2dF) as well as very deep, high-redshift, ones (e.g. Hubble Deep Field, GROTH, DEEP2), carried out by means of the new generation of very large ground-based telescopes and sophisticated detectors on board of satellites covering the whole electromagnetic spectrum, from gamma to radio wavelengths.

Gaia and the Galaxy

How did the first galaxies form and evolve? How they distribute around the Universe? These are, among others, long standing questions aimed to understand the process of galaxy formation in the Universe. ESA's satellite Gaia will significant contribute to this challenge by providing the ultimate map of the sky. For certain, its catalogues will underpin pretty much all of astronomy for decades to come. **Credit:** ESA.

Lines of Research

Semianalytical modeling of galaxy formation.

Dark matter clustering and halo structure and kinematics.

Galaxy evolution in groups and clusters.

Kinematics and structure of the galaxy.

Members

Abedi, Hoda; Balaguer-Núñez, M. Dolores; Carrasco, José M.; Castañeda, Javier; Czekaj, Maria A.; Darriba, Laura; Fabricius, Claus V.; Figueras, Francesca; Jordi, Carme; Luri, F. Xavier; Masana, Eduard; Monguió, Maria; Palmer, Max; Roca-Fàbrega, Santiago; Romero-Gómez, Mercè; Solanes, José M.; Toribio, M. Carmen; Torra, Jordi; Voss, Holger; Weiler, Michael.

Gravitation and Cosmology

The ICCUB's researchers carry out research in the areas of gravity, particle physics and the gauge/gravity correspondence. In the area of gravity, research is focused on the formulation of effective theories for black holes, inflationary models and quantum gravity in de Sitter spaces. In particle physics, the main topic is phenomenological implications of supersymmetric extensions of the Standard Model. Finally, the gauge/gravity correspondence is applied to the study of the quark-gluon plasma and the computation of observables in gauge theories.

Lines of Research

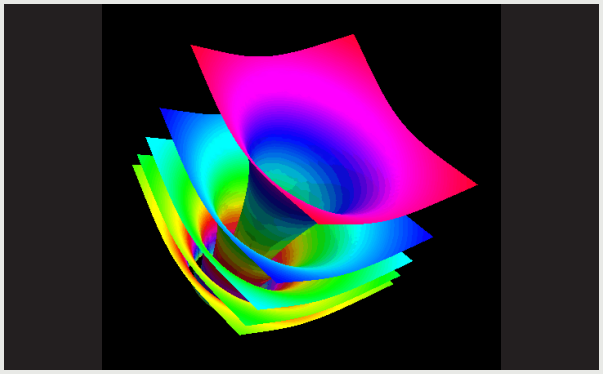
Dark matter and dark energy in cosmology and in particle physics.

Quantum and semiclassical gravity.

Black holes and gravity: New perspectives from strings and higher dimensions.

Members

Di Dato, Adriana; Cámara, Pablo G.; Emparan, Roberto A.; Fernández, Daniel; Fiol, Bartomeu; Fröb, Markus; Garolera, Blai; Garriga, Jaume; Guasch, Jaume; Haddad, Nidal; Llosa, Josep; Martínez, Marina; Mateos Solé, David; Molina, Alfred; Notari, Alessio; Peñaranda, Siannah; Solà, Joan; Tanabe, Kentaro; Tarrio, Luis Javier; Torrents, Genís; Urakawa, Yuko; Verdaguer, Enric.



Curved spacetime: from black holes to cosmology, from quarks to quantum gravity

Einstein's theory of gravity tells us that the geometry of space and time gets dramatically distorted in the vicinity of black holes and at the beginning of the universe. This challenges the foundations of Einstein's theory itself and calls for the incorporation of the effects of quantum mechanics. A surprising spin-off of this research is the possibility of describing a ball of quark-gluon plasma as a black hole in a higher-dimensional space.

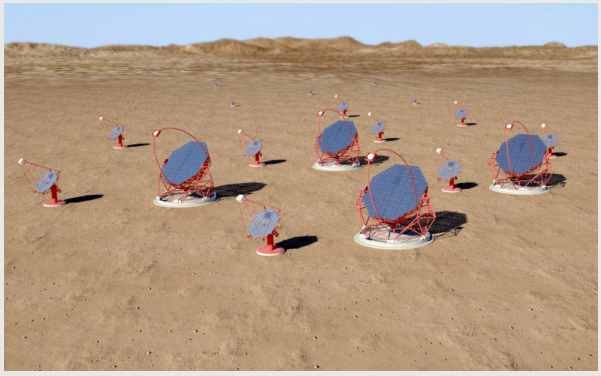
High Energy Astrophysics

The general aim of the ICCUB's researchers working on this field is to achieve a better understanding of relativistic flows in three archetypical classes of sources: microquasars, gamma-ray binaries, pulsar wind nebulae and active galactic nuclei, focusing on their study at high energies. These sources are archetypical in the sense that they resemble in many aspects other types of objects (some of them non-relativistic), like young stellar objects with jets, massive star binaries with colliding winds, or gamma ray bursts. This aim will be attained by gathering data over a large wavelength range (from radio to TeV energies), as well as by modeling emission processes in different scenarios (jets, shocks, interaction with the interstellar medium, etc.).

Moreover, the ICCUB's high energy astrophysicists are members of the MAGIC Collaboration since February 2006, and are now participating, together with experimental physicists and engineers of the ICCUB, in the Cherenkov Telescope Array (CTA) project. CTA is an initiative to build the next generation ground-based very high energy gamma-ray instrument. It will serve as an open observatory to a wide astrophysics community and will provide a deep insight into the non-thermal high-energy universe. In addition to physics case studies, the ICCUB is involved in the design and prototyping of CTA telescope cameras. This activity involves designing ASICs (Application Specific Integrated Circuits) for pre-amplification, amplification, signal processing and triggering.

Members

Bosch-Ramon, Valentí; Gascón, David; Iwasawa, Kazushi; Marcote, Benito; Migliari, Simone; Moldón, Fco. Javier; Munar-Adrover, Pere; Paredes-Fortuny, Xavier; Paredes, Josep Maria; Ribó, Marc; Sanuy Charles, Andreu; Zanin, Roberta.



Cherenkov Telescope Array (CTA)

CTA is an initiative to build the next generation ground-based instrument for very-high-energy (VHE) gamma ray astronomy, ICCUB researchers are involved both scientifically and technically in the design of the project. **Credit:** G Pérez/IAC/SMM.

Lines of Research

High-energy and very-high-energy gamma-ray sources in the Galaxy.

Multi-wavelength observations and theoretical modeling.

Microquasars.

Gamma-ray binaries.

Pulsar wind nebulae.

Active galactic nuclei.

MAGIC and Cherenkov Telescope Array.

Nuclear and Hadronic Physics

Nuclear physicists at the ICCUB are actively participating in studies of neutron-rich nuclei, a research which is closely related to that conducted on the topics of nuclear equation of state and its astrophysical applications. They have also benefited from the computational power offered by the most modern supercomputers (JLab, Fermilab, Tungsten and Mare Nostrum), which has allowed them to perform large dynamical simulations related to their studies.

Lines of Research

Hadronic physics. Strangeness and charm in the nuclear medium.

Lattice QCD of low-energy hadronic interactions.

Nuclear structure. Nuclear symmetry energy.

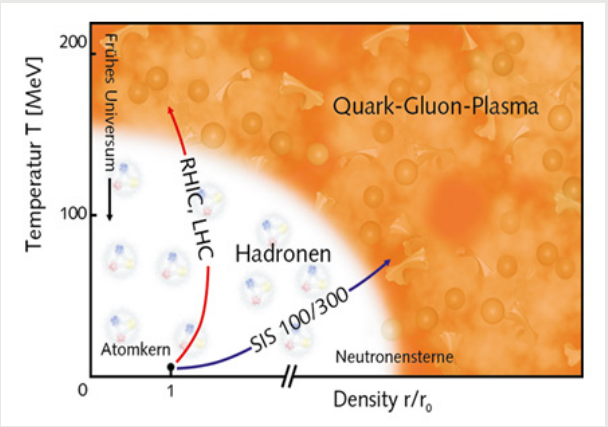
Relativistic heavy ion collisions.

Dense and hot nuclear matter and applications in nuclear astrophysics.

Radiation transport and interactions of radiation with matter.

Members

Carbone, Arianna; Centelles, Mario; Chang, Emmanuel Zhi Yin; Magas, Volodymyr; Mateos, David; Parreño, Assumpta; Pérez-Obiol, Axel; Polls, Artur; Ramos, Àngels; Sharma, Bharat Kishore; Viñas, Xavier.



Phase diagram of hadronic matter as predicted by theory

The diagram plots the temperature in units of one million electronvolt against the density in units of normal nuclear density ρ_0. At very high temperatures and densities, physicists expect that the quarks and their bonding particles, the gluons - normally locked up inside the nucleons - become liberated from their confinement and move as free particles in a so-called quark-gluon plasma.
Credit: FAIR@GSI.

Particle Physics Phenomenology

The ICCUB has a wide spectrum of interests in the phenomenological and calculational aspects of particle physics, including many aspects of the areas reported in the hep-ph and hep-th archives.

Recently, its activity has been influenced to a large extent by the commissioning of the LHC. In this sense, the studies are being focused on effective theories of the symmetry breaking sector of the Standard Model, some aspects of supersymmetric theories, string phenomenology, flavour physics (particularly b-physics) and physics beyond the standard model that the LHC will continue to explore in years to come.

The ICCUB's researchers are also active in heavy-quark effective theories and other effective theories of QCD. Several key features of heavy ion collisions and the properties of QCD under extreme conditions are also receiving attention.

This area also includes work on lattice field theory, particularly in connection with b-physics.

The ICCUB's researchers have relevant activity in the development of parton distribution functions using neural networks.

The scope of future accelerators in the context of some extensions of the Standard Model are studied by researchers at the ICCUB.

Relevant contributions are also being made in the field of neutrino physics, axion physics and other dark matter candidates, and dark energy.

The ICCUB's researchers of this area have close interactions with the experimental particle physicists at the ICCUB and obviously with researchers in other theoretical areas.

Members

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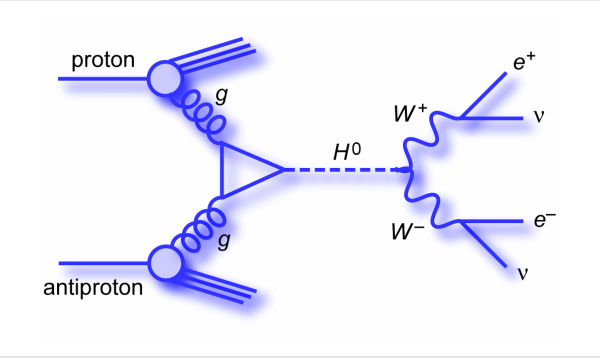


Feynman diagram for production and decay of a Higgs boson

A Feynman diagram is a pictorial representation of mathematical expressions governing the behaviour of subatomic particles. The interaction of subatomic particles can be complex and difficult to understand intuitively, and the Feynman diagrams allow a simple visualization of what would otherwise be a rather abstract formula. **Credit:** Ann Heinson.

Lines of Research

Standard Model and beyond at the LHC.

B-physics, with an emphasis on the analysis and physical reach of the LHCb detector.

Phenomenology of supersymmetric theories.

String phenomenology.

Unification of the fundamental forces.

Heavy quark effective theory and other effective theories of QCD.

QCD in extreme conditions: heavy ion experiments at the LHC, FAIR and other accelerators.

Lattice QCD.

Perturbative QCD: parton distribution functions.

Studies of the physics of future colliders.

Physics of neutrinos, with an emphasis on astrophysics and cosmology.

Axions and other dark matter candidates.

This is a miscellaneous group formed by small independent subunits devoted to the development of new scientific tools and technological applications useful for or arisen from the research of other ICCUB groups, or that could develop in new consolidated groups in the general domain of space sciences and quantum physics.

Lines of Research

Astrodynamics and celestial mechanics.

Astronomical image processing and high angular resolution techniques.

Chirality and prebiotic chemistry.

Data processing and analysis.

Heliospheric physics and space weather.

Instrumentation.

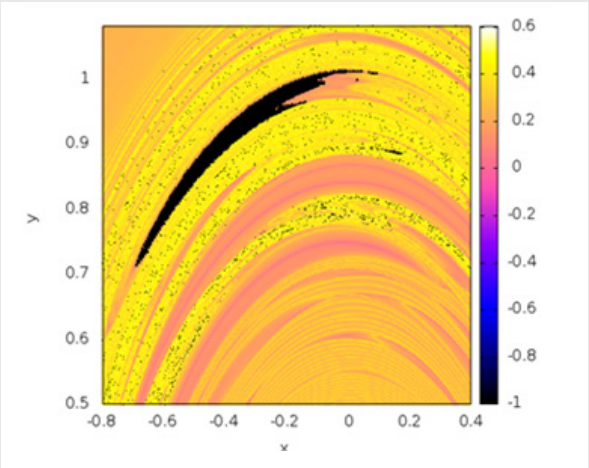
Microgravity.

Astrodynamics and Celestial Mechanics

The ICCUB's researchers on Astrodynamics are devoting their efforts to addressing some fundamental issues concerning formation flying for multiple spacecraft. These include: the transfer of a set of spacecraft to either an Earth orbit, or to a libration zone, the deployment of formations from stacks of satellites and the proximity manoeuvring for pointing and reconfiguration.

The main goal of the ICCUB's researchers is to develop and implement algorithms based on recent advances obtained by them and other collaborating teams. The proposed methodology will enable the transfer of spacecraft to tactical locations by developing a strategy that mimics those of flocks of birds.

Further developments will incorporate collision avoidance algorithms currently under development in the area of complexity science, that have evolved from the principles of molecular dynamics. The implications of developing these methodologies are far reaching and could potentially impact on path planning methodologies throughout the physical sciences.



Coherent Lagrangian structures

The Coherent Lagrangian Structure formalism (LCS) can be used to study the dynamics of the three body problem.

Members

Gómez, Gerard; Olikara, Zubin Philip; Paita, Fabrizio; Perez-Palau, Daniel; Yijun, Lian.

Lines of Research

Develop tools to explain in a natural way different astronomical patterns.

Astronomical Image Processing and High Angular Resolution Techniques

The ICCUB's researchers in the field of image reconstruction are focused on exploiting the use of the wavelet transform to improve the ability of image sensors to detect faint stars and moving objects. The effects of the curvelet transform over interferometric images are also being studied, differential photometry is being estimated in adaptive optics observations using a wavelet-based maximum likelihood estimator.

The ICCUB's researchers are also working on obtaining super-resolution using additive-substitutive wavelets techniques on remotely sensed images, as well as in obtaining new high-sensitivity, milliarcsecond resolution results from observations of lunar occultations at Very Large Telescope (VLT) of the European Southern Observatory (ESO).

Lines of Research

Image deconvolution by means of multiresolution analysis (wavelet transform).

Image fusion by means of multiresolution analysis (wavelet and curvelet transforms).

Image superresolution by means of multiresolution analysis (wavelet transform).

Submilliarcsecond resolution of infrared sources by high time resolution lunar occultations technique.

Members

Baena, Roberto; Fors, Octavi; Merino, M. Teresa; Núñez, Jorge C.

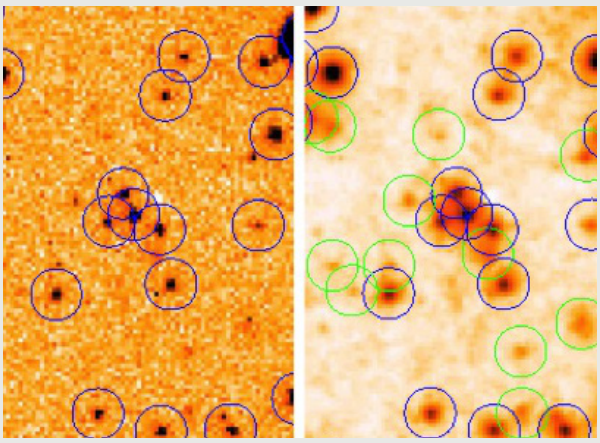


Image reconstruction

Image reconstruction makes substantial improvements in qualitative appearance and recovery of faint star images in terms of the number of stars that can be detected. In the raw frame (left) of a CCD exposure obtained with a Baker-Nunn camera, stars detected by the Sextractor algorithm are circled in blue. In the restored frame (right) the additionally detected stars are circled in green. The wavelet based AWMLE algorithm was used.

Chirality and Prebiotic Chemistry

Researchers at ICCUB have focused on the implementation of Mueller matrix methods in order to study the emergence of chirality by the application of a gradient of shear rates by flows in different chemical systems. Theoretical and experimental methods are studied in the deracemization and spontaneous emergence of chirality in crystallizations. Moreover, the ICCUB's researchers have proved mechanical chiral fields should be included among the few physical chiral forces that can induce chirality at the primordial scenarios (astrophysical or planetary) of the chemical evolution.

Lines of Research

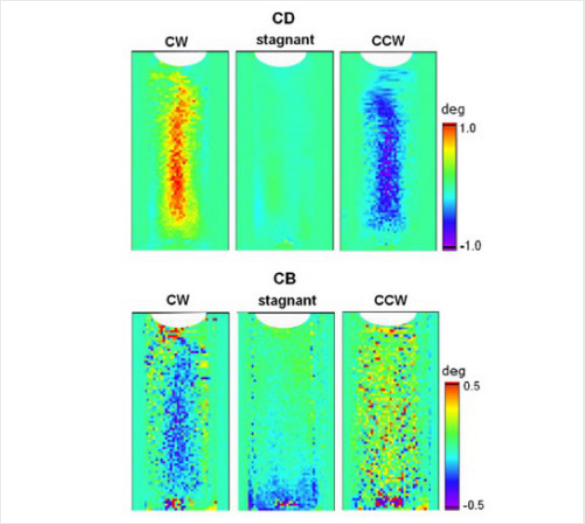
Effect of mechanical forces (flows with gradient of shear rates) on the emergence of chirality in soft matter.

Mirror symmety breaking in crystallizations and aggregations showing critical phenomena.

Organocatalysis in asymmetric synthesis.

Members

Crusats, Joaquim; Ribó, Josep M.; Ríos, Ramon; El-Hachemi, Zoubir.



Chirality

Scanning of CD and CB (natural optical activity) inside a 10 mm pathlenght cuvette for clockwise (CW), counter-clockwise (CCW) and stagnant (no stirring). CD and CB have been calculated by ICCUB researchers from space-resolved measurements of the Mueller matrix performed in situ at 485 nm, that corresponds to a peak at high energy of the CD bisignated CD band.

Data Processing and Analysis

The ICCUB's researchers are engaged since 1998 in the Gaia Data Processing and Analysis Consortium (DPAC) in charge of designing, implementing, managing and running the whole data reduction of the Gaia mission, from the storage of the telemetry to the production of the final catalogue.

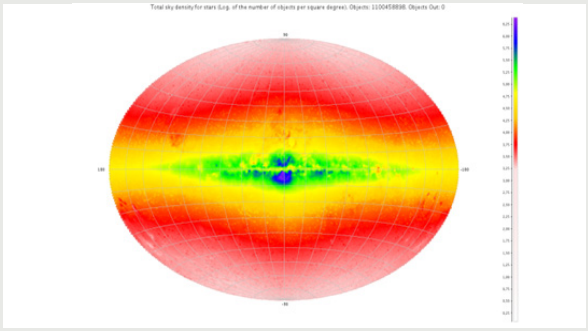
The ICCUB has important responsibilities in four out of nine coordination units in DPAC: CU2 (simulations), CU3 (core processing), CU5 (photometric processing), CU9 (Catalogue Access) and in the Data Processing Center of Barcelona (comprising BSC and CESCA).

In this framework, the ICCUB also leads the EU FP7 funded initiative named GENIUS (2013-2016), aimed to significantly contribute to the development of the Gaia Archive: use the best state-of-the-art archive system; provision of exploitation tools to maximize the scientific return; ensuring the interoperability with future astronomical archives; and last but not least, the archive facilities outreach activities. Our team has one representative in the Gaia Scientific Team, one in the DPAC Executive, two deputy managers in CU2 and CU3, and leads the CU9.

Based on the expertise of our team on efficient compression systems for space, DAPCOM was created as a spin-off company dedicated to efficient data compression systems.

Members

Antiche, Erika; Balaguer-Núñez, M. Dolores; Borrachero, Raul; Carrasco, José M.. Castañeda, Javier; Clotet, Marcial; Fabricius, Claus V.; Fries, Aidan Dermot; Gallardo, Eva; Garralda, Nora; González, Juan José; Jordi, Carme; Julbe, Francesc; Luri, F. Xavier; Masana, Eduard; Molina, Daniel; Portell, Jordi; Sagristà, Antoni; Torra, Jordi; Voss, Holger; Weiler, Michael.



Simulation of the Milky Way as seen by Gaia

ESA's Gaia astrometric mission (2013-2018) will be the most accurate optical astronomy satellite ever built so far. It will create a census of a billion stars, finding clues of the origin, structure and evolution of our own Galaxy, the Milky Way. The development of the Gaia archive is one of the most demanding European Grand Challenge of the next decade.

Lines of Research

Data reduction of space missions.

Data compression.

Data mining.

Software engineering.

Parallel processing & massive data.

Heliospheric Physics and Space Weather

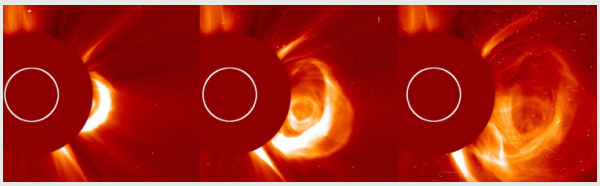
The ICCUB's lines of research of in heliophysics mainly deal with SEP events triggered by solar activity and interplanetary disturbances, i.e. energetic protons (~150 keV to 0.5 GeV) and near relativistic electrons (~20 keV up to 0.5 MeV). Solar flares and coronal mass ejections, which are the main agents of SEP-acceleration, together with proxies of solar activity (radioemission, H-alpha, X-ray, etc.), and the solar wind plasma and the interplanetary magnetic field, are also topics of research at the ICCUB, as are background components of the SEP scenario.

Similarly, the ICCUB's researchers are working on data analysis and the study of SEP events, individual cases and multispacecraft events (ACE, Wind, Ulysses, SOHO, ISEE-3, Helios, Goes, IMPs, Phobos, STEREO and other spacecraft and satellites), multi-instrument observations and interdisciplinary analysis of relevant solar events, from Sun to Earth magnetosphere. They are also modeling gradual proton events, and giving scientific support to the participation of technological groups of the UB in ESA's Solar Orbiter project: Polarimetric and Helioseismic Imager (PHI) and LET-EPD (Energetic Particle Detector) instruments.

Finally, space weather tools for the prediction of SEP intensity-time profiles are also being developed.

Members

Àgueda, Neus; Aran, Àngels; Sanahuja, Blai.



Coronal mass ejection

On 2012 May 17 at 01:48 UT, a coronal mass ejection erupted from the west limb of the Sun (Active Region 1746) at speed of 1582 km/s. This eruption was accompanied by an M-class flare and produced a solar energetic particle event.
Credit: ESA & NASA/SOHO.

Lines of Research

Solar energetic particle events, interplanetary shocks and related solar activity.

Modeling gradual proton events: magneto hydrodynamics (MHD) shock simulation plus particle transport simulation.

Modeling near-relativistic electron events: inversion methods.

Space weather: Engineering models for solar energetic particle events.

Within the context of instrumentation, the ICCUB participates very actively in:

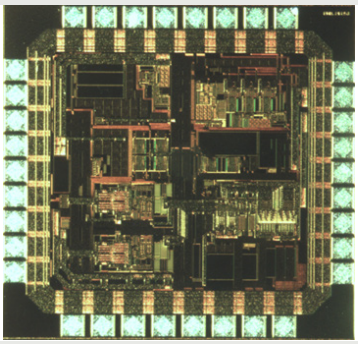
A positioner for SIDE, MEGARA and BIG-BOSS
The ICCUB has been responsible for the development of the electronics for the fiber positioner built by the AVS Company (Spain). This positioner has had successive prototypes starting with the one for the SIDE/GTC, up to the MEGARA/GTC (3th generation) MOS spectrograph. Both will operate in the Gran Telescopio de Canarias (GTC). The positioner has also been proposed for the BIG-BOSS Spectrograph (NOAO-USA).

Deployable probe arms for MIRADAS
MIRADAS is a 3rd generation near-infrared multi-object echelle spectrograph working at spectral resolution of 20.000 for the GTC which is expected to start operations in 2016. The ICCUB, as member of the MIRADAS consortium, is in charge of the design of the deployable probe arms with pickoff mirror optics.

Improvement of the TFRM (Fabra-ROA Telescope)
The ICCUB is involved in the improvement of the telescope, which has now begun observing Extrasolar planets, Space debris (ESA program), near-Earth objects (NEOs), Optical counterparts of gamma ray bursts (GRBs) and some X-Ray binaries (microquasars).

Elaboration of the Science Case document of WEAVE
The ICCUB has participated in the elaboration of the Science Case document of WEAVE, a multi-object spectrograph for the William Herschell Telescope (Canary Islands) expected to be approved in 2013 and operational by the end of 2016. This instrument will provide complementary on-ground spectroscopic observations for the Gaia mission (ESA).

Design of radiation-tolerant Application-Specific Integrated Circuits (ASICs) for CTA, LHCb upgrade and PET
The CTA project is an initiative to build the next generation ground-based very high energy gamma-ray instrument. In addition to physics case studies, the ICCUB is involved in the design and prototyping of CTA telescope cameras. This activity involves designing ASICs (Application Specific Integrated Circuits) for preamplification, signal processing and triggering). On the other hand, the ICCUB is also working in the design of an ASIC for the calorimeter of the upgraded LHCb and the development of ASICs for new PET (Positron Emission Tomography) systems based in silicon photomultipliers.



ASICs for CTA

Low noise and high dynamic range preamplifier for photomultiplier tubes:
- 500 MHz bandwidth
- 16 bits dynamic range
- 1 bi-gain channel
- Tecnology: AMS SiGe BiCMOS 0.35µm
- Area: 2.4 mm2

Projects

A Positioner for SIDE, MEGARA and BIG- BOSS.

Deployable probe arms for MIRADAS.

Improvement of the TFRM (Fabra-ROA Montsec Telescope).

Elaboration of the Science Case document of WEAVE.

Design of radiation-tolerant Application- Specific Integrated Circuits (ASICs) for CTA, LHCb upgrade and PET.

Members

Casajús, Adrià; Comerma-Montells, Albert; Figueras, Francesca; Fors, Octavi; Garrido, Lluís; Gascón, David; Gómez, Jose M.; Graciani Díaz, Ricardo; Graugés, Eugeni; Jordi, Carme; Lazovski, Nikola; Núñez, Jorge C.; Picatoste Olloqui, Eduard; Ruiz, Hugo; Sabater, Josep; Sanuy, Andreu; Torra, Jordi; Trenado, Juan; Vilar, Cristian.

Microgravity and Biphasic Fluxes

The ICCUB's collaborators have focused on the formation and management of small bubbles under microgravity conditions, an area of fundamental interest in two phase-flows, with important applications in space technology, from life support systems to thermal control of satellites. Experiments are conducted in the ESA Drop Tower facility at ZARM (Bremen) and include the study of the interaction of bubbles and turbulence, and the formation of vapor bubbles by nucleate boiling. Researchers have obtained for the first time turbulent monodisperse bubble suspensions, and also for the first time, bubbles of controlled size by nucleate boiling.

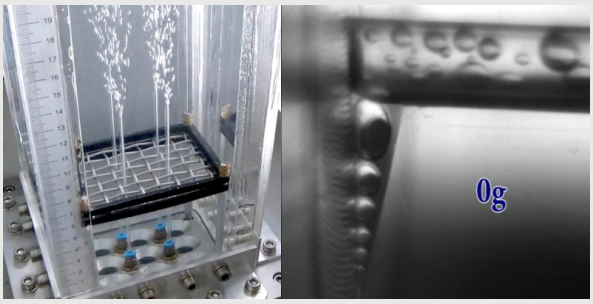
Lines of Research

Bubble formation and dynamics in turbulent flows in microgravity.

Controled nucleate boiling in confined geometries in microgravity.

Invited Researchers

Casademunt, Jaume; Ruiz, Josep Xavier.



Bubbles in microgravity

Details of two setups used in Drop Tower experiments to study bubble formation and dynamics in turbulent flows and controlled boiling in confined geometries.

Star Formation

The ICCUB's main research topic in the field of star formation is the detailed investigation of the first stages of stellar evolution in star-forming regions in our Galaxy. The ICCUB's researchers intend to acquire a perspective as wide as possible by including observations which range from the optical to the radio domain, and integrating the results within complete and self-consistent theoretical models, in what constitutes a double approach, theoretical and observational.

Star formation represents a challenge in observational techniques, since it takes place in high-density cores in the molecular clouds of the Galaxy, where extinction is very high. It is, furthermore, also a challenge to theoretical astrophysics. While the formation of an isolated low-mass star is a process well understood, there is still ample controversy on the mechanisms that allow to accrete the mass required to form a star of more than 10 times the mass of the Sun.

The ICCUB's researchers have been able to observe, through the polarization of the emission, the small-scale magnetic field around protostars, and the results obtained point to the magnetic field as responsible for controlling the onset of gravitational collapse of dense molecular cores.

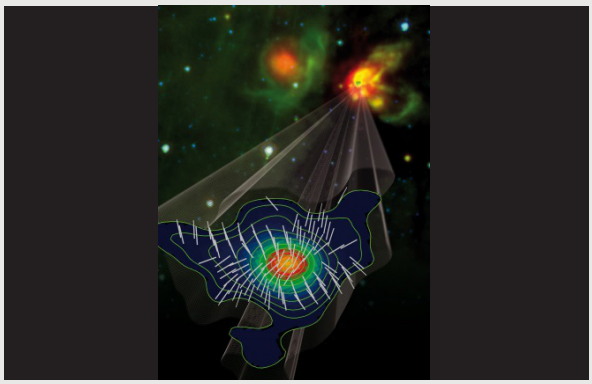
Other areas of research in this field include the chemical evolution of the molecular gas, the study of the launching and collimation of the astrophysical jets associated with young stellar objects and planetary nebulae; the investigation of the early stages of the formation of massive stars; the characterization of the gravitational collapse that drives the full process of star formation and the search for signatures of planet formation within the protoplanetary disks; and the study of the transition from hot molecular cores to bright HII regions.

Members

Estalella, Robert; Frau, Pau; López, Rosario; Masqué, Josep M.; Padoan, Paolo.

Invited Researchers

Torrelles, Josep M.



G31.41+0.31

Contour map of the 879-μm dust emission superposed on the color image of the polarized flux intensity in units of Jy per beam. Black thick bars indicate the position angle of the magnetic field. These maps were obtained by using a natural weighting to the visibility data, which yielded to a full width at half maximum synthesized beam of 1.34" x 0.83" with a position angle of 67°. **Credit:** Girart, J.M., Beltrán, M.T., Zhang, Q., Rao, R., Estalella, R.

Lines of Research

High-angular resolution observations of the first stages of stellar evolution.

Outflows, jets, and accretion disks in low- and high-mass young stellar objects.

Jets in planetary nebulae.

Computational models of star-forming clouds and star formation.

Theoretical Physics

ICCUB's activities cover an ample spectrum of the areas reported in the hep-th and quant-ph archives.

String theory has inspired in recent times enormous activity in the gauge/string duality conjecture that allows a treatment of some strongly coupled theories in terms of a gravity dual. These techniques are being applied to the study of the quark-gluon plasma. In this domain of physics, techniques such as the gauge/string duality meet nuclear phenomenology and effective theories and there is substantial cross-fertilization.

Duality techniques are also applied to a variety of strongly coupled condensed matter systems and in problems of quantum information.

Supersymmetric field theories are studied seeking to understand the ultraviolet behavior of theories with extended supersymmetries and deriving exact results.

In another line, some theories of emergent gravity have been proposed. There are are also obvious connections with cosmology and the physics of black holes. Research in this area share many common interests with the Gravitation and Cosmology one.

The research in quantum information can be roughly divided into six topics: multiparticle entanglement; low energy properties of many-body quantum systems; quantum error correction; topological order; ultra-cold gases; and quantum simulation.

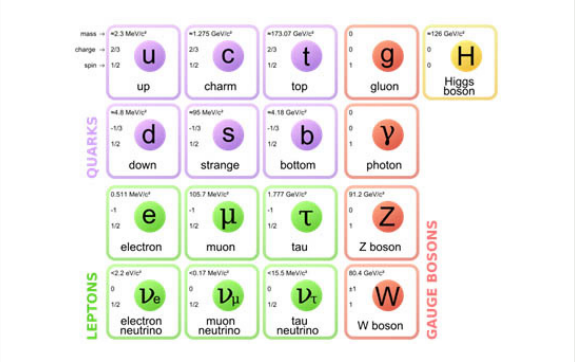
The quantum information researchers at the ICCUB are in close collaboration with some of the groups at ICFO.

Members

Aprile, Francesco; Barranco, Alejandro; Boada, Octavi; Cámara, Pablo G.; Casalderrey- Solana, Jorge; Dector, Aldo; Espriu, Domènec; Fernández, Daniel; Fiol, Bartomeu; Forini, Valentina; Garolera, Blai; Gomis, Joaquim; Iblisdir, Sofyan; Latorre, José I; Mariño, Mauricio; Mateos, David; Pablos, Daniel; Pons, Josep M.; Puigdomènech, Daniel; Russo, Jorge G.; Solà, J.; Tarrio, Luis J.; Tywoniuk, Konrad.

Invited Researchers

Lizzi, Fedele; Talavera, Pere.



Standard model of elementary particles

The Standard Model includes 12 fundamental fermions, 4 fundamental bosons and the postulated Higgs boson. However, it is not complete: why quarks and leptons have such diferent values? Why there exists three families of quarks and leptons? Why has the photon no mass and Z boson does? Where does dark matter fit into the model? **Credit:** Wikipedia.

Lines of Research

String theory.

Supersymmetric field theories.

Applications of the gauge/string duality to QCD.

Application of the gauge/string duality to condensed matter systems.

Low energy properties of many-body quantum systems.

Quantum error correction.

Topological order.

Ultra-cold gases.

Quantum simulations.

PROJECTS AND FUNDS

National Plan Projects

Astrophysics and Space Sciences

Modelos experimentales y teóricos sobre la emergencia de quiralidad
Reference: AYA2009-13920-C02-02
PI: Albert Moyano, UB (ICCUB: Josep M.. Ribó)
Agency: MICINN
Duration: 2010-2012
Amount: 136.000 €

Fluctuaciones primordiales y fisica de cuásares con 100.000 espectros de cuásares
Reference: AYA2009-09745
PI: Jordi Miralda-Escudé
Agency: MICINN
Duration: 2009-2012
Amount: 70.000 €

Las componentes del universo
Reference: AYA2009-13667
PI: M. Pilar Ruiz-Lapuente
Agency: MINECO
Duration: 2010-2012
Amount: 87.120 €

Propiedades de la luz intragrupal y de las galaxias en grupos galaxias
Reference: AYA2010-18605
PI: José M. Solanes
Agency: MICINN
Duration: 2011-2013
Amount: 35.695 €

Contribución al desarrollo científico y tecnológico de la misión Gaia
Reference: AYA2009-14648-C02-01
PI: Jordi Torra
Agency: MICINN
Duration: 2010-2013
Amount: 2.757.342 €

High-energy phenomena in stellar objects. Theory and multi-wavelength observations
Reference: AYA2010-21782-C03-01
PI: Josep M. Paredes
Agency: MICINN
Duration: 2011-2013
Amount: 129.470 €

Participación española en la fase de preparación del "Cherenkov Telescope Array" (CTA)
Reference: FPA2010-22056-C06-02
PI: Marc Ribó
Agency: MICINN
Duration: 2011-2013
Amount: 290.400 €

CTA, el Cherenkov Telescope Array, una instalación avanzada para la astronomía gamma desde tierra
Reference: EUI2009-04072
PI: Manuel Martínez, IFAE (ICCUB: Josep M. Paredes)
Agency: MICINN
Duration: 2009-2012
Amount: 405.000 € (ICCUB: 200.000 €)

Optimización del retorno científico de la observación astronómica. Nuevos desarrollos y aplicaciones
Reference: AYA2008-01225
PI: Jorge Núñez
Agency: MICINN
Duration: 2009-2013
Amount: 85.000 €

Explotación del modelo AMIGA de formación y evolución de galaxias
Reference: AYA2009-12792-C03-01
PI: Eduard Salvador-Solé
Agency: MICINN
Duration: 2010-2012
Amount: 92.444 €

Auto-organización en materiales blandos y materia viva: II) Fluidos complejos, células y tejidos
Reference: FIS2010-21924-C02-02
PI: Jaume Casademunt
Agency: MICINN
Duration: 2011-2013
Amount: 217.800 €

Sucesos de partículas solares energéticas: modelos. Aplicaciones para meteorología espacial
Reference: AYA2010-17286
PI: Blai Sanahuja
Agency: MICINN
Duration: 2011-2013
Amount: 266.200 €

Diseño detallado de SOLAR ORBITER/PHI
Reference: AYA2011-29833-C06-05
PI: Josep M. Gómez
Agency: MINECO
Duration: 2012-2013
Amount: 640.090 €

Interstellar medium at high-angular resolution: preparing for the ALMA era
Reference: AYA2011-30228-C03-03
PI: Robert Estalella
Agency: MINECO
Duration: 2012-2014
Amount: 45.000 €

Cosmology and the Origin of Matter. Sabor y Origen de la Materia (COM SOM)
Reference: FPA2011-29678-C02-02
PI: Licia Verde
Agency: MINECO
Duration: 2012-2014
Amount: 148.000 €

Nuclear and Particle Physics and Gravitation

Mantenimiento y operación de Tier2 español para LHCb y contribuciones al core computing de LHCb
Reference: FPA2010-21885-C02-01
PI: Ricardo Graciani Díaz
Agency: MICINN
Duration: 2011-2013
Amount: 211.629 €

Desarrollo de nuevos detectores para los futuros colisionadores en Física de Partículas
Reference: FPA2010-21549-C04-01
PI: Ángel Diéguez
Agency: MICINN
Duration: 2011-2013
Amount: 459.800 €

Estudio de la violación de CP con el detector LHCb
Reference: FPA2011-30163-C02-01
PI: Eugeni Graugés
Agency: MICINN
Duration: 2012-2014
Amount: 655.820 €

Teoría y fenomenología de las interacciones fundamentales: Gravitación y cosmología
Reference: FPA2010-20807-C02-02
PI: Roberto Emparan
Agency: MICINN
Duration: 2011-2013
Amount: 213.000 €

Información cuántica: entrelazamiento, redes de tensores y gases fríos
Reference: FIS2010-16185
PI: José Ignacio Latorre
Agency: MICINN
Duration: 2011-2014
Amount: 187.550 €

Simulación Monte Carlo del transporte de radiación. Física, métodos numéricos y aplicaciones
Reference: FPA2009-14091-C02-01
PI: Francesc Salvat
Agency: MICINN
Duration: 2010-2013
Amount: 61.500 €

Teorías efectivas de las interacciones fuertes: aplicaciones a quarkonium pesado y a QCD bajo condiciones extremas
Reference: FPA2010-16963
PI: Joan Soto
Agency: MICINN
Duration: 2011-2013
Amount: 112.409 €

Teoría y fenomenología de las interacciones fundamentales: Física de partículas y la unificación de las fuerzas
Reference: FPA2010-20807-C02-01
PI: Domènec Espriu
Agency: MICINN
Duration: 2011-2013
Amount: 422.169 €

Sistemas cuánticos en interacción: hadrones, núcleos y átomos
Reference: FIS2008-01661
PI: Assumpta Parreño
Agency: MICINN
Duration: 2009-2012
Amount: 338.074 €

Sistemas de Fermi fuertemente correlacionados: átomos, núcleos y hadrones
Reference: FIS2011-24154
PI: Xavier Viñas
Agency: MICINN
Duration: 2012-2014
Amount: 204.490 €

Special and Complementary Actions

Astrophysics and Space Sciences

Creación de la Red Española Gaia
Reference: AYA2009-08488-E
PI: Francesca Figueras
Agency: MICINN
Duration: 2010-2012
Amount: 38.000 €

Jornadas de impulso a la explotación científica de Gaia
Reference: AYA2011-15133-E
PI: Francesca Figueras
Agency: MICINN
Duration: 2011-2012
Amount: 5.900 €

Ampliación de actividades de la misión GAIA
Reference: AYA2010-12176-E
PI: Jordi Torra
Agency: MICINN
Duration: 2011-2013
Amount: 184.000 €

Fenómenos de alta energía en eyecciones relativistas
Reference: AYA2010-09310-E
PI: Josep M. Paredes
Agency: MICINN
Duration: 2011-2012
Amount: 12.000 €

Preparación del IPDR de SO/Phi por parte de la UB
Reference: AYA2010-09789-E
PI: Josep M. Gómez
Agency: MICINN
Duration: 2011-2012
Amount: 20.000 €

Participación española en la fase preparatoria del Cherenkov Telescope Array (CTA)
Reference: AIC-A-2011-0660
PI: Manuel Martínez, IFAE (ICCUB: Josep M.Paredes)
Agency: MINECO
Duration: 2011-2014
Amount: 432.677 € (ICCUB: 0 €)

La contribución de las ICTS españolas a la misión Gaia de ESA
Reference: MICINN-RIA
PI: Carme Jordi
Agency: MICINN
Duration: 2012
Amount: 8.800 €

Nuclear and Particle Physics and Gravitation

Desintegraciones raras e identificación del sabor de mesones B en LHCb
Reference: AIC-D-2011-0681
PI: Hugo Ruiz
Agency: MICINN
Duration: 2011-2012
Amount: 3.300 €

Ecuación de estado de materia nuclear asimetrica
Reference: AIC10-D-000608
PI: Artur Polls
Agency: MICINN
Duration: 2010-2012
Amount: 2.900 €

Desarrollo de la matriz densidad e interacciones nucleares efectivas para los campos Hartree-Fock y de apareamiento
Reference: AIC10-D-000592
PI: Xavier Viñas
Agency: MICINN
Duration: 2010-2012
Amount: 4.550 €

Participación en el Computing resources Scrutiny Group del CERN
Reference: FPA2011-13440-E
PI: Domènec Espriu
Agency: MICINN
Duration: 2011-2013
Amount: 9.000 €

Ruptura de simetría en fisica de partículas: el Higgs y más allá
Reference: AIC-D-2011-0815
PI: Domènec Espriu
Agency: MICINN
Duration: 2012-2013
Amount: 3.000 €

Año dual España-Rusia: Física de Partículas, Física Nuclear y Astropartículas
Reference: FPA2011-14321-E
PI: Domènec Espriu
Agency: MICINN
Duration: 2011-2012
Amount: 8.000 €

Conferencia Internacional "International Conference on Hypernuclear and Strange Particle Physics"
Reference: FIS2011-15579-E
PI: Àngels Ramos
Agency: MICINN
Duration: 2012-2013
Amount: 9.000 €

National Plan Consolider-Ingenio Projects

Astrophysics and Space Sciences

Supercomputación y eCiencia
Reference: CSD2007-00050
PI: Mateo Valero, BSC
(ICCUB: Jordi Torra)
Agency: MEC
Duration: 2007-2013
Amount: 5.000.000 € (ICCUB: 150.000 €)

Primera ciencia con el GTC: la astronomía española en vanguardia de la astronomía europea
Reference: CSD2006-00070
PI: José Miguel Rodríguez Espinosa, IAC
(ICCUB: Eduard Salvador)
Agency: MEC
Duration: 2007-2012
Amount: 5.000.000 € (ICCUB: 0 €)

Nuclear and Particle Physics and Gravitation

Centro Nacional de Física de Partículas, Astropartículas y Nuclear
Reference: CSD2007-00042
PI: Antonio Pich, IFIC
(ICCUB: Lluís Garrido)
Agency: MEC
Duration: 2007-2012
Amount: 10.000.000 € (ICCUB: 479.200 €)

Canfranc Underground Physics
Reference: CSD2008-00037
PI: M. Concepción González-García
Agency: MICINN
Duration: 2008-2013
Amount: 5.000.000 € (ICCUB: 470.250 €)

Consolidated Groups

Astrophysics and Space Sciences

Maximizing the scientific return of future galaxy surveys
Reference: 2009SGR1280
PI: Licia Verde
Agency: AGAUR
Duration: 2009-2014
Amount: 42.640 €

Astronomia i Astrofisica
Reference: 2009SGR217
PI: Eduard Salvador-Solé
Agency: AGAUR
Duration: 2009-2013
Amount: 58.240 €

Nuclear and Particle Physics and Gravitation

Grup de Física Experimental d'Altes Energies
Reference: 2009SGR1268
PI: Lluís Garrido
Agency: AGAUR
Duration: 2009-2013
Amount: 47.840 €

Gravitation, Particles and Strings
Reference: 2009SGR168
PI: David Mateos
Agency: AGAUR, Generalitat de Catalunya
Duration: 2009-2013
Amount: 44.720 €

Dosimetria i Radiofisica Mèdica
Reference: 2009SGR0276
PI: Francesc Salvat
Agency: AGAUR
Duration: 2009-2013
Amount: 46.800 €

Grup de Física Teòrica d'Altes Energies
Reference: 2009SGR502
PI: Joan Solà
Agency: AGAUR
Duration: 2009-2013
Amount: 58.240 €

Laboratori de física matemàtica
Reference: 2009SGR417
PI: Josep Llosa
Agency: AGAUR
Duration: 2009-2013
Amount: 87.500 €

European Projects and Funds

Astrophysics and Space Sciences

Cosmological physics with future large scale structure surveys (PHYSS.LSS)
Reference: 240117 (FP7-IDEAS-ERC)
PI: Licia Verde
Agency: European Research Council (ERC)
Duration: 2009-2014
Amount: 1.395.000 €

Gaia Research for European Astronomy Training (GREAT)
Reference: 08-RNP-118
PI: Nick Walton, University of Cambridge
(ICCUB: Carme Jordi)
Agency: European Science Foundation (ESF)
Duration: 2010-2015
Amount: 735.000 €

Gaia Research for European Astronomy Training (GREAT-ITN)
Reference: 264895 (FP7-PEOPLE)
PI: Nick Walton (University or Cambridge;
ICCUB: Francesca Figueras)
Agency: European Community (EC)
Duration: 2011-2015
Amount: 4.250.580 €
(ICCUB: 456.239 €)

WGA1 Gaia model workshop: Galaxy modelling with a Gaia mock catalogue
Reference: PESC-3857
PI: Daisuke Kawata, UCL (ICCUB: Francesca Figueras)
Agency: European Science Foundation (ESF)
Duration: 2011-2012
Amount: 7.560 €

The Gaia DPAC Interface Management in the Gaia Project Office
Reference: GAIA-CT-12000-178-CN
PI: Jordi Torra
Agency: Centre National d'Études Spatiales (CNES)
Duration: 2009-2013
Amount: 320.000 €

The Preparatory Phase for the Cherenkov Telescope Array (CTA-PP)
Reference: 262053 (FP7-INFRASTRUCTURES)
PI: Werner Hofmann, Max Planck Gesellschaft
(ICCUB: Josep M. Paredes)
Agency: European Community (EC)
Duration: 2010-2013
Amount: 5.200.000 €

Star Formation in the Turbulent Interstellar Medium
Reference: PIRG07-GA-2010-261359 (FP7- PEOPLE)
PI: Paolo Padoan, Eduard Salvador-Solé

Agency: European Community (EC)
Duration: 2011-2014
Amount: 100.000 €

Data Services and Analysis Tools for Solar Energetic Particle Events and Related Electromagnetic Emissions (SEPserver)
Reference: 2010.2.1-03 (FP7-SPACE)
PI: Rami Vainio, University of Helsinki
(ICCUB: Blai Sanahuja)
Agency: European Community (EC)
Duration: 2010-2013
Amount: 1.932.175 €
(ICCUB: 150.562,50 €)

Protecting Space Assets from High Energy Particles by Developing European Dynamic Modelling and Forecasting Capabilities (SPACECAST)
Reference: SPA. 2010.2.3-01 (FP7-SPACE)
PI: Richard Horne, British Antartic Survey
(ICCUB: Blai Sanahuja)
Agency: European Community (EC)
Duration: 2011-2014
Amount: 1.965.076 € (ICCUB: 160.004 €)

SSA - CO-VI Optical Observations For Space Surveillance And Tracking Test And Validations (TFRM)
Reference: 4000107443/12/D/MRP
PI: Jorge Núñez
Agency: European Space Agency (ESA)
Duration: 2012-2013
Amount: 50.000 €

Demonstration Test-Bed for the Remote Control of an Automated Follow-Up Telescope (TFRM participation)
Reference: G532-004GR
PI: I. Dominguez, Ground Station department, INSA
(ICCUB: Jorge Núñez)
Agency: European Space Agency (ESA)
Duration: 2012-2014
Amount: 800.000 € (ICCUB: 38.000 €)

The Astrodynamics Network (ASTRONET-II)
Reference: PITN-GA-2011-289240 (FP7-PEOPLE)
PI: Gerard Gómez
Agency: European Community (EC)
Duration: 2012-2015
Amount: 3.888.318 €

Nuclear and Particle Physics and Gravitation

Holography for the LHC era Reference (HoloLHC)
Reference: 306605 (FP7-IDEAS-ERC)
PI: David Mateos
Agency: European Research Council (ERC)
Duration: 2012-2017
Amount: 1.419.424 €

Advanced European Infrastructures for Detectors at Accelerators (AIDA)
Reference: 262025 (FP7-INFRASTRUCTURES)
PI: Ángel Diéguez
Agency: European Community (EC)
Duration: 2011-2015
Amount: 8.000.000 € (ICCUB: 44.920 €)

Study of Strongly Interacting Matter (HADRONPHYSICS3)
Reference: INFRA-2011-1-1-20 (283286)
(FP7- INFRASTRUCTURES)
PI: Carlo Guaraldo
(INFN Frascati; ICCUB: Àngels Ramos)
Agency: European Community (EC)
Duration: 2012-2014
Amount: 9.000.000 € (ICCUB: 19.000 €)

Multi-hadron interactions in Lattice QCD: 27,1 Mh in Mare Nostrum
Reference: MHILQCD
PI: Assumpta Parreño
Agency: Partnership for advanced computing in Europe (PRACE)
Duration: 2012-2013
Amount: 27.100.000 CPUh

Light quark mass dependence of two-hadron energies in Lattice QCD: 30 Mh at CURIE Thin node partition
Reference: 2011040561
PI: Assumpta Parreño
Agency: Partnership for Advanced Computing in Europe (PRACE)
Duration: 2012
Amount: 30.000.000 CPUh

Standard Model and New Physics with the LHCb detector (SM-NewPhysics-LHCb)
Reference: PERG04-GA-2008-235071 (FP7- PEOPLE)
PI: Domenèc Espriu, David d'Enterria

Agency: European Community (EC)
Duration: 2009-2012
Amount: 45.000 €

European Particle Physics Latin America Network (EPLANET)
Reference: PIRSES-GA-2009-246806 (FP7- PEOPLE)
PI: Luciano Maiani, CERN (ICCUB: Domènec Espriu)
Agency: European Community (EC)
Duration: 2011-2015
Amount: 3.245.400 €
(ICCUB: 59.400 €)

INVISIBLES
Reference: PITN-GA-2011-289442 (FP7-PEOPLE)
PI: B. Gavela, UAM
(ICCUB: M. Concepción González García)
Agency: European Community (EC)
Duration: 2012-2016
Amount: 3.245.400 € (ICCUB: 152.566 €)

International Projects

Astrophysics and Space Sciences

Multi-spacecraft Observations of Near- Relativistic Electron Events: modeling their solar injection and interplanetary transport
Reference: NXX09AG30G
PI: David Lario, APL. Johns Hopkins University
(ICCUB: Neus Àgueda)
Agency: NASA
Duration: 2009-2012
Amount: 199.926 €
(ICCUB: 0 €)

Preconditioning of the interplanetary medium as responsible for large intense SEP events: Radial and longitudinal effects
Reference: NNX11AO83G
PI: David Lario,
APL. Johns Hopkins University
(ICCUB: Neus Àgueda)
Agency: NASA
Duration: 2011-2015
Amount: 404.081 €
(ICCUB: 0 €)

Contract for the preliminary design of the MIRADAS Spectograph Probe Motion Control Software System for the Gran Telescopio Canarias
Reference: MIRADAS Contract
PI: Jordi Torra
Agency: University of Florida
Duration: 2012-2013
Amount: 73.125 €

Other Funds and Contracts

Astrophysics and Space Sciences

Atorgament d'un ajut de la convocatòria del Programa d'Incentivació de la Intensificació de l'activitat investigadora 2012.
Reference: PIRSES-GA-2009-246806
PI: Blas Sanahuja
Agency: UB Duration: 2012-2013
Amount: 4.973,10 €

Análisis de la Integración de un Sensor Radar Doppler en el asiento de un vehículo
Reference: FBG 306487
PI: Josep M. Gómez
Company: Fico Mirrors, S.A.
Duration: 2011-2012
Amount: 37.850 €

Interplanetary and Planetary Radiation Model for Human Spaceflight
Reference: FBG 307130
PI: Daniel Heynderickx
(ICCUB: Àngels Aran, Blai Sanahuja)
Company: DH Consultancy/ESA
Duration: 2012-2013
Amount: 150.000 € (ICCUB: 9.000 €)

Ajuts per finançar la participació en projectes internacionals de recerca (2011-2012): GENIUS
PI: Xavier Luri
Agency: Universitat de Barcelona

Duration: 2011-2012
Amount: 4.500 €

Injection of nucleate-boiling slug flows into a heat exchange chamber in microgravity
Reference: FA8655-12-1-2060
PI: Jaume Casademunt
Agency: Air Force Office of Scientific Research (Dept. of Defense USA) through the European Office of Aerospace Research and Development
Duration: 2012-2015
Amount: 179.540 €

Estancias de profesores e investigadores extranjeros de acreditada experiencia en régimen de año sabático en centros españoles
Reference: SAB2010-0120
PI: Jordi Torra, Luís Aguilar (UNAM)
Agency: MEC
Duration: 2011-2012
Amount: 31.900 €

PTA Mod. Infraestructuras científico-técnicas: Gaia
Reference: PTA2010-3704-I
PI: Jordi Torra
Agency: MINECO
Duration: 2011-2014
Amount: 56.700 €

El tránsito de Venus 2012 y las distancias cósmicas
Reference: FCT-11-2316
PI: Carme Jordi
Agency: FECYT
Duration: 2011-2012
Amount: 4.000 €

El Trànsit de Venus 2012 i la divulgació de l'astronomia a la Universitat de Barcelona
Reference: 2012ACDC00161
PI: Carme Jordi
Agency: AGAUR
Duration: 2012-2012
Amount: 5.000 €

Eclipsi 2.0: una aplicació Android per al càlcul d'eclipsis i trànsits planetaris
Reference: 2012ACDC00081
PI: Jordi Torra
Agency: AGAUR
Duration: 2012-2012
Amount: 6.000 €

Nuclear and Particle Physics and Gravitation

Miniaturization of the controller for a endoscopic screening capsule, Phase 4
Reference: FBG 307059
PI: Ángel Diéguez
Company: Ovesco Endoscopy AG
Duration: 2012
Amount: 9.920 €

Asesoramiento sobre el desarrollo y prueba de circuitos electrónicos para discriminación de señales en detectores de partículas
Reference: FBG 306720
PI: Lluís Garrido
Company: Scientifica Internacional
Duration: 2012-2014
Amount: 11.000 €

Analysis with Penelope
Reference: FBG 306890
PI: Francesc Salvat
Company: Hamamatsu Photonics K.K.
Duration: 2012-2013

Intellectual services relative to support to Penelope training course
Reference: FBG 30704
PI: Francesc Salvat
Agency: Organisation for economic co-operation and development
Duration: 2012
Amount: 4.000 €

Miniaturization of the controller for an endoscopic screening capsule, Phase 2
Reference: FBG 306675
PI: Ángel Diéguez
Agency: Ovesco Endoscopy AG
Duration: 2011-2012
Amount: 82.088 €

Miniaturization of the controller for an endoscopic screening capsule, Phase 3
Reference: FBG 306988
PI: Ángel Diéguez
Agency: Ovesco Endoscopy AG
Duration: 2012
Amount: 14.880 €

PUBLICATIONS

SCI Publications

Astrophysics and Space Sciences

Abramowski, A.; et al. (ICCUB: **Cañellas, A.; Moldón, J.; Munar-Adrover, P.; Paredes, J.M.; Ribó, M.; Zabalza, V.; Zanin, R.**), "The 2010 Very High Energy γ-Ray Flare and 10 Years of Multi-wavelength Observations of M 87", Astrophysical Journal, vol. 746, p. 151 (2012).

Àgueda, N.; Lario, D.; Ontiveros, V.; Kilpua, E.; **Sanahuja, B.**; Vainio, R., "Multi-spacecraft Study of the 8 November 2000 SEP Event: Electron Injection Histories 100° Apart", Solar Physics, vol. 281, num. 1, p. 319-331 (2012).

Àgueda, N.; Vainio, R.; **Sanahuja, B.**, "A Database of >20 keV Electron Green's Functions of Interplanetary Transport at 1 AU", Astrophysical Journal Supplement Series, vol. 202, num. 2 (2012).

Ahn, C.P.; et al. (SDSS-III Collaboration; ICCUB: **Verde, L.; Miralda-Escudé, J.**), "The Ninth Data Release of the Sloan Digital Sky Survey: First Spectroscopic Data from the SDSS-III Baryon Oscillation Spectroscopic Survey", Astrophysical Journal Supplement Series, vol. 203, p. 21 (2012).

Alba, A.N.R.; Valero, G.; Calbet, T.; Font-Bardía, M.; Moyano, A.; **Ríos, R.**, "Enantioselective addition of oxazolones to maleimides. An easy entry to quaternary aminoacids", New Journal of Chemistry, vol. 36, p. 613 - 618 (2012).

Aleksic, J. et al. (MAGIC Collaboration; ICCUB: **Bosch-Ramon, V. ; Cañellas, A.; Moldón, J.; Munar-Adrover, P.; Paredes, J.M.; Ribó, M.; Zabalza, V.; Zanin, R.**), "Detection of the γ-Ray Binary LS I +61°303 in a Low-flux State at Very High Energy γ-Rays with the MAGIC Telescopes in 2009", Astrophysical Journal, vol. 746, p. 80 (2012).

Aleksic, J. et al. (MAGIC Collaboration; ICCUB: **Bosch-Ramon, V.; Cañellas, A.; Moldón, J.; Munar-Adrover, P.; Paredes, J.M.; Ribó, M.; Zabalza, V.; Zanin, R.**), "Detection of VHE γ-Rays from HESS J0632+057 during the 2011 February X-Ray Outburst with the MAGIC Telescopes", Astrophysical Journal Letters, vol. 754L, p. 10A (2012).

Aleksic, J. et al. (MAGIC Collaboration; ICCUB: **Bosch-Ramon, V.; Cañellas, A.; Moldón, J.; Munar-Adrover, P.; Paredes, J.M.; Ribó, M.; Zabalza, V.; Zanin, R.**), "PG 1553+113: five years of observations with MAGIC", Astrophysical Journal, vol. 748, p. 46A (2012).

Aleksic, J. et al. (MAGIC Collaboration; ICCUB: **Bosch-Ramon, V.; Cañellas, A.; Moldón, J.; Munar-Adrover, P.; Paredes, J.M.; Ribó, M.; Zabalza, V.; Zanin, R.**), "Phase-resolved energy spectra of the Crab pulsar in the range of 50-400 GeV measured with the MAGIC telescopes", Astronomy & Astrophysics, vol. 540, p. 69A (2012).

Aleksic, J. et al. (MAGIC Collaboration; ICCUB: **Cañellas, A.; Moldón, J.; Munar-Adrover, P.; Paredes, J.M.; Ribó, M.; Zabalza, V.; Zanin, R.**), "Detection of very-high energy γ-ray emission from NGC 1275 by the MAGIC telescopes", Astronomy & Astrophysics, vol. 539, p. L 2A (2012).

Aleksic, J. et al. (MAGIC Collaboration; ICCUB: **Cañellas, A.; Moldón, J.; Munar-Adrover, P.; Paredes, J.M.; Ribó, M.; Zabalza, V.; Zanin, R.**), "Discovery of VHE γ-ray emission from the BL Lacertae object B3 2247+381 with the MAGIC telescopes", Astronomy & Astrophysics, vol. 539A, p. 118A (2012).

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Aleksic, J. et al. (MAGIC Collaboration; ICCUB: **Cañellas, A.; Moldón, J.; Munar-Adrover, P.; Paredes, J.M.; Ribó, M.; Zabalza, V.; Zanin, R.**), "High zenith angle observations of PKS 2155-304 with the MAGIC-I telescope", Astronomy & Astrophysics, vol. 544A, p. 75A (2012).

Aleksic, J. et al. (MAGIC Collaboration; ICCUB: **Cañellas, A.; Moldón, J.; Munar-Adrover, P.; Paredes, J.M.; Ribó, M.; Zabalza, V.; Zanin, R.**), "MAGIC observations of the giant radio galaxy M 87 in a low-emission state between 2005 and 2007", Astronomy & Astrophysics, vol. 544A, p. 96A (2012).

Aleksic, J. et al. (MAGIC Collaboration; ICCUB: **Cañellas, A.; Moldón, J.; Munar-Adrover, P.; Paredes, J.M.; Ribó, M.; Zabalza, V.; Zanin, R.**), "Morphological and spectral properties of the W51 region measured with the MAGIC telescopes", Astronomy & Astrophysics, vol. 541A, p. 13A (2012).

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Anderson, L.; et al. (ICCUB: **Verde, L.**), "The clustering of galaxies in the SDSS-III Baryon Oscillation Spectroscopic Survey: Baryon Acoustic Oscillations in the Data Release 9 Spectroscopic Galaxy Sample", MNRAS, vol. 427, p. 3435-3467 (2012).

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Casares, J.; **Ribó, M.**; Ribas, I.; **Paredes, J. M.**; Vilardell, F.; Negueruela, I., "On the binary nature of the γ-ray sources AGL J2241+4454 (= MWC 656) and HESS J0632+057 (= MWC 148)", MNRAS, vol. 421, p. 1103 (2012).

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Collins, D.C.; Kritsuk, A.G.; **Padoan, P.**; Li, H.; Xu, H.; Ustyugov, S.D.; Norman, M.L., "The Two States of Star-forming Clouds", Astrophysical Journal, vol. 750, num. 1, p. 13 (2012).

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Salgado, C.A., **D'Enterria, D.**; et al., "Proton-Nucleus Collisions at the LHC: Scientific Opportunities and Requirements", J.Phys. G, vol. 39, p. 015010 (2012)

Das, S.; **De Putter, R.**; V. Linder, E.; Nakajima, R., "Weak lensing cosmology beyond ACDM", Journal of Cosmology and Astroparticle Physics, vol. 2012, num. 11, p. 011 (2012).

De Putter, R.; Wagner, Ch.; Mena, O.; **Verde, L.**; Percival, W., "Thinking Outside the Box: Effects of Modes Larger than the Survey on Matter Power Spectrum Covariance", JCAP, vol. 1204, p. 019 (2012).

De Putter, R.; Mena, O.; Giusarma, E.; Ho, S.; Cuesta, A.; Seo, H.J.; Ross, A.J.; White, M.; Bizyaev, D.; Brewington, H.; Kirkby, D.; Malanushenko, E.; Malanushenko, V.; Oravetz, D.; Pan, K.; Percival, W.J.; Ross, N.P.; Schneider, D.P.; Shelden, A.; Simmons, A.; Snedden, S., "New neutrino mass bounds from SDSS-III data release 8 photometric luminous galaxies", Astrophysical Journal, vol. 761, num. 1, p. 12 (2012).

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"The star-forming core ahead of HH 80N: studying the interaction with a parsec scale jet"
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"The role of magnetic fields in the formation of low and high mass stars"
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Supervisor/s: Josep M. Girart, M. Teresa Beltrán
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Supervisor/s: Marc Ribó, Josep M. Paredes
Defense Date: 05/07/2012

"Turbulent bubble suspensions and crystal growth in microgravity. Drop tower experiments an numerical simulations"
Author: Pau Bitlloch
Supervisor/s: Jaume Casademunt, Josep Xavier Ruiz
Defense Date: 11/10/2012

"The Evolution, Masses and Morphologies of Merging"
Author: Kevin Casteels
Supervisor/s: Eduard Salvador-Solé
Defense Date: 19/10/2012

"A new version of the Besançon Galaxy Model constrained with Tycho data"
Author: Maria Anna Czekaj
Supervisor/s: Annie C. Robin, Francesca Figueras, Xavier Luri
Defense Date: 22/10/2012

"Energía Oscura y Gravedad Modificada"
Author: Miguel Zumalacarregui
Supervisor/s: Juan García-Bellido, Pilar Ruiz-Lapuente, Tomi S. Koivisto
Defense Date: October 2012

"AMIGA: Parallelization, Lyα Emission Line And Steps Towards WDM Implementation"
Author: Jordi Viñas
Supervisor/s: Alberto Manrique
Defense Date: 23/11/2012

Nuclear and Particle Physics and Gravitation

"Charmed Baryon Resonances and mesons in Hot and Dense matter"
Author: Clara Estela Jiménez
Supervisor/s: Àngels Ramos, Isaac Vidaña
Defense Date: 05/03/2012

"First measurements of radiative B decays in LHCb"
Author: Albert Puig Navarro
Supervisor/s: Ricardo Graciani
Defense Date: 09/03/2012

"Enabling active locomotion and advanced features in an endoscopic capsule"
Author: Oscar Alonso
Supervisor/s: Angel Diéguez
Defense Date: 27/04/2012

"Flavour Tagging developments within the LHCb experiment"
Author: Marc Grabalosa
Supervisor/s: Marco Musy
Defense Date: 15/05/2012

"Explicit Bound States and Ressonaces fields in Effective Field Theories"
Author: Jaume Tarrús
Supervisor/s: Joan Soto, Pere Talavera
Defense Date: 07/06/2012

"Studies with Massive Neutrinos in Particle Physics, Astrophysics and Cosmology"
Author: Jordi Salvadó
Supervisor/s: M. Concepción González-García
Defense Date: 27/09/2012

"Black holes: New perspectives from higher-dimensions"
Author: Nidal Haddad
Supervisor/s: Roberto Emparan
Defense Date: 29/10/2012

"Simulació quàntica amb reticles òptics"
Author: Octavi Boada
Supervisor/s: José I. Latorre
Defense Date: 02/11/2012

"Spontaneous generation of geometry and its possible consequences"
Author: Daniel Puigdomènech
Supervisor/s: Domènec Espriu
Defense Date: 06/11/2012

"Measurement of the direct CP asymmetry in $B^0 \to K^* \gamma$ decays with the LHCb detector"
Author: Ricardo Vázquez -Gómez
Supervisor/s: Ricardo Graciani, Hugo Ruiz
Defense Date: 12/12/2012

Master Theses

Astrophysics and Space Sciences

"Imaging of star formation regions with AIPS"
Author: Carmen Juárez
Supervisor/s: Robert Estalella
Defense Date: 01/02/2012

"The quasar - Lyman alpha cross-correlation in BOSS"
Author: Eduard Arnau
Supervisor/s: Jordi Miralda
Defense Date: 29/06/2012

"Expected contents of the Gaia catalogue for binary stars"
Author: Núria Vinyoles
Supervisor/s: Xavier Luri
Defense Date: 07/09/2012

"Low-frequency radio observations of gamma-ray binaries"
Author: Benito Marcote
Supervisor/s: Josep Maria Paredes
Defense Date: 07/09/2012

Nuclear and Particle Physics and Gravitation

"On extremal black holes in the Einstein Yang-Mills system"
Author: Luís Cort
Supervisor/s: Bartomeu Fiol
Defense Date: 08/02/2012

"Stability, Thermodynamics and Critical Phenomena in Black Strings"
Author: Marina Martínez
Supervisor/s: Roberto Emparan
Defense Date: 08/02/2012

"Non-Gaussian features of primordial fluctuations in single field inflationary models"
Author: Mariona Anglada
Supervisor/s: Jaume Garriga
Defense Date: 02/07/2012

"Inflation from a Higgs false vacuum: Post-Inflationary evolution"
Author: Ramon Nogueira
Supervisor/s: Alessio Notari
Defense Date: 02/07/2012

"Improving vertex fitting with a photon"
Author: Damián Álvarez
Supervisor/s: Lluís Garrido
Defense Date: 20/09/2012

"Entanglement entropy as an order parameter: a holographic perspective"
Author: Daniel Alsina
Supervisor/s: Bartomeu Fiol
Defense Date: 20/09/2012

"Radiative Energy Loss in a Thin QCD Medium"
Author: Daniel Pablos
Supervisor/s: Jorge Casalderrey
Defense Date: 20/09/2012

"WW amplitudes in a effective theory"
Author: Emilio Flores
Supervisor/s: Domènec Espriu, Joan Soto
Defense Date: 20/09/2012

"Applications of Matrix Product States"
Author: Javier García
Supervisor/s: José I. Latorre
Defense Date: 20/09/2012

"Modelos evolutivos de la energía de vacío en cosmología"
Author: Pello Bilbao
Supervisor/s: Joan Solà
Defense Date: 20/09/2012

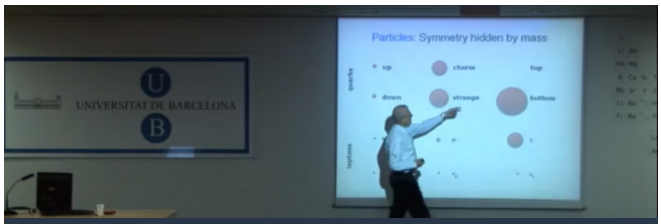
"Aspects of Anisotropic Plasmas"
Author: Teresa García
Supervisor/s: David Mateos
Defense Date: 20/09/2012

"Reacción N→KΞ en modelos quirales con canales acoplados hasta next-to-leading order"
Author: Albert Feijoo
Supervisor/s: Volodymyr Magas
Defense Date: 06/11/2012

ACTIVITIES

ICCUB Colloquia

Close, Frank (Rudolf Peierls Centre for Theoretical Physics)
"The Infinity Puzzle: from atoms to the LHC and how to win a Nobel Prize - but did you deserve it?"
23/01/2012



ICCUB Colloquium given by F. Close the 23th January 2012

Badenes, Carles
(University of Pittsburgh, Pennsylvania, USA)
"Type Ia Supernova Progenitors: Ripple in Still Water"
05/03/2012

Protopapas, Pavlos
(Harvard-Smithsonian Center for Astrophysics)
"Interdisciplinary science"
12/03/2012

Salgado, Carlos
(Universidad de Santiago de Compostela)
"Hot and dense QCD in the LHC era"
19/03/2012

Perna, Rosalba (University of Colorado at Boulder)
"Gamma-Ray Bursts as Tools for Extragalactic Astrophysics and Cosmology"
26/03/2012

Juste, Aurelio (IFAE, Institut de Física d'Altes Energies)
"The Higgs boson discovery: a solution to a massive problem"
12/09/2012

Mahmoudi, Nazila (CERN Theory division)
"Flavour Physics in the LHC era"
17/09/2012

Rajagopal, Krishna (MIT)
"The Hottest, and most Liquid, Liquid in the Universe"
22/10/2012

Seminars

ICCUB Seminars

S. Eikenberry, Stephen
(Department of Astronomy. University of Florida)
"The Next-Generation Infrared Spectrograph for the GTC"
06/03/2012

Verdes-Montenegro, Lourdes
(Instituto de Astrofísica de Andalucía)
"The Square Kilometer Array: a challenge for ~ 2020 to which Spain can contribute in 2012"
11/04/2012.

Gil de Paz, Armando
(Universidad Complutense de Madrid)
"MEGARA, the next-generation optical spectrograph of GTC"
24/04/2012

Aguilar, Luís (UNAM-UB)
"Stable regions around stars in eccentric binary systems"
16/05/2012

Guzmán, Rafael (University of Florida)
"Proyecto científico-tecnológico de micro/nano satélites"
24/05/2012

Campos Plasencia, Isabel
(Instituto de Fisica de Cantabria - IFCA)
"Ibergrid: Una Infraestructura de Computación al servicio de los investigadores"
06/06/2012

Haugboelle, Troels (Center for Star and Planet Formation - STARPLAN, University of Copenhagen)
"Large Scale Models of Star-Forming Regions"
15/06/2012

Clarkson, Chris (Cape Town University)
"Dark Energy and Inhomogeneity"
19/06/2012

Arnau, Eduard (ICCUB)
"Modelling the QSO-LYalpha cross-correlation"
29/06/2012

Saito, Shun (Univ. Berkeley)
"Non-linear redshift space distortions in the angular power spectrum"
17/07/2012

Reid, Beth (Berkeley U.)
"SDSS-III Baryon Oscillation Spectroscopic Survey DR9 Results: baryon acoustic oscillations, the growth of structure, and the Alcock-Pacyznski effect at z=0.57"
18/07/2012

Djordjevic, Goran (University of Nis, Serbia)
"On Nonlocal Inflation, Tachyons and (Non) Minimal Coupling"
27/07/2012

Miralda, Jordi (ICCUB)
"The Practice of Theoretical Astrophysics"
18/10/2012

Font, Andreu (University of Zurich)
"Cosmology with the Lyman alpha Forest: First results from the BOSS survey"
12/11/2012

Group Seminars

Mannarelli, Massimo (INFN Gran Sasso)
"Superluminal neutrinos and pion decay"
HEP Seminar
12/01/2012

Puigdomènech, Daniel (ICCUB)
"Spontaneous generation of geometry in four dimensions"
HEP Seminar
13/01/2012.

Entem, David R. (U. Salamanca)
"Nuclear Forces in Chiral EFT"
FAN Seminar
18/01/2012

Quartin, Miguel (U. Federal do Rio de Janeiro)
"Interesting Anisotropic and Inhomogeneous Cosmologies"
HEP Seminar
19/01/2012

Racker, Juan (ICCUB)
"Leptogenesis with small violation of B-L"
HEP Seminar
20/01/2012

Anabalon, Andrés (U. Adolfo Ibanez, Chile)
"Asymptotically AdS Black Holes and Wormholes with a Self Interacting Scalar Field in Four Dimensions"
HEP Seminar
26/01/2012

Lizzi, Fedele (Naples U.)
"Spectral action, scale anomaly and the Higgs-Dilaton potential"
HEP Seminar
27/01/2012

Molina, Raquel (IFIC, Valencia)
"A new interpretation for the Ds2(2573), the prediction of novel exotic charmed mesons and narrow N*, Lambda* resonances around 4.3 GeV "
FAN Seminar
01/02/2012

G. Cámara, Pablo (ICCUB)
"Flavor physics from local F-theory GUTs"
HEP Seminar
03/02/2012

Mateu, Cecilia (CIDA. Mérida, Venezuela)
"Adventures in the Milky Way: From RR-Lyrae and the Thick Disk to Gaia and Stellar Streams in the Halo"
DAM Seminar
08/02/2012

Booth, Ivan (UB & Memorial U.)
"Near Equilibrium Black Holes and Branes"
HEP Seminar
09/02/2012

Forini, Valentina (ICCUB)
"Quark-antiquark potential in AdS/CFT"
HEP Seminar
10/02/2012

Vernizzi, Filippo (CEA Gif-sur-Yvette)
"Squeezing the CMB and the cosmic shear bispectra"
HEP Seminar
14/02/2012

Notari, Alessio (ICCUB)
"Predicting the Higgs mass from Inflation"
HEP Seminar
16/02/2012

Aprile, Francesco (ICCUB)
"AdS/Condensed Matter Physics. A first example: Holographic Superconductors"
HEP Seminar
17/02/2012

Ortin, Tomás (UAM)
"Non-extremal black holes and branes of N=2, d=4,5 Supergravity"
HEP Seminar
23/02/2012

Yencho, Brian (ICCUB)
"Azimuthal Correlations in Top Pair Decays and the Effects of New Heavy Scalars"
HEP Seminar
24/02/2012

Ramallo, Alfonso (U. de Santiago de Compostela)
"Holographic flavors and impurities in Chern-Simons-matter theories"
HEP Seminar
01/03/2012

Andrianov, Alexander
(ICCUB & St. Petersburg State U.)
"Flying of vector particles from a parity breaking medium to vacuum and back"
HEP Seminar
02/03/2012

Pineda, Antonio (UAB)
"The muonic hydrogen lamb shift and the proton radius"
HEP Seminar
08/03/2012

Garolera, Blai (ICCUB)
"Exact results for static and radiative fields of a quark in N=4 super Yang-Mills"
HEP Seminar
09/03/2012

Zabalza, Victor (Max-Planck-Institut für Kernphysik, Heidelberg, Alemanya.)
"A tale of two components: Revealing the origin of the high energy gamma-ray emission from LS 5039."
DAM Seminar
12/03/2012

Bernabeu, José (U. Valencia & IFIC)
"Time reversal violation from the entangled B0-0 system"
HEP Seminar
15/03/2012

Urakawa, Yuko (ICCUB)
"Infrared stability of de Sitter spacetime"
HEP Seminar
16/03/2012

Van der Schee, Wilke (Utrecht U.)
"Strong coupling isotropization simplified"
HEP Seminar
22/03/2012

Barranco, Alejandro (ICCUB)
"N=1 SQCD-like theories with Nf massive flavors from AdS/CFT and beta functions"
HEP Seminar
23/03/2012

Tremblay, Pier-Emmanuel (LSW, Heidelberg)
"The high-mass problem in cool white dwarfs: an old obstacle now seen with 3D radiation-hydrodynamics "
DAM Seminar
28/03/2012

Gaberdiel, Matthias (Zurich ETH)
"Minimal Model Holography"
HEP Seminar
28/03/2012

Espriu, Domènec (ICCUB)
"Gravitational waves in the presence of a cosmological constant"
HEP Seminar
30/03/2012

Verdes-Montenegro, Lourdes (IAA-CSIC)
"The Square Killometer Array: a challenge for ~ 2020 to which Spain can contribute in 2012."
DAM Seminar
11/04/2012

Landsteiner, Karl (IFT-UAM)
"Anomalous Transport and Kubo Formulae"
HEP Seminar
12/04/2012

Casalderrey, Jorge (ICCUB)
"Off-diagonal Flavour Susceptibilities from AdS/CFT"
HEP Seminar
20/04/2012

Segovia, Jorge (U. Salamanca)
"Heavy meson phenomenology in a
constituent quark model"
FAN Seminar
25/04/2012

Milhano, J. G.
(CENTRA Lisboa & CERN)
"Colour matters: the importance of medium
modification of colour flow to jet quenching"
HEP Seminar
26/04/2012

Niro, Viviana (ICCUB)
"Indirect detection of Dark Matter
with neutrino detectors"
HEP Seminar
27/04/2012

Soto, Joan (ICCUB)
"Heavy quarkonium in a moving thermal bath"
HEP Seminar
04/05/2012

Furuya, Ray S. (Subaru Telescope, National
Astronomical Observatory of Japan)
"The Initial Conditions for Gravitational
Collapse of a Low-Mass Star-Forming Core"
DAM Seminar
10/05/2012

Rahatlou, Shahram
(U. Roma "La Sapienza" & INFN Roma)
"Exotica at Large Hadron Collider"
HEP Seminar
10/05/2012

Puig, Albert (ICCUB)
"Radiative B decays in LHCb"
HEP Seminar
11/05/2012

Csernai, Laszlo P.
(U. Bergen, Norway)
"Quark Gluon Plasma search:
from heavy ion collisions to neutron stars"
HEP Seminar
15/05/2012

Shinnaga, Hiroko
(California Institute of Technology
Submillimeter Observatory)
"Magnetic Field in the Isolated Massive Dense Clump
IRAS 20126+4104"
DAM Seminar
17/05/2012

Gomis, Joaquim (ICCUB)
"Non-linear Realizations, Goldstone bosons of broken
Lorentz rotations and effective actions for p-branes"
HEP Seminar
18/05/2012

Muga, Juan Gonzalo (U. País Vasco)
"How to speed up quantum adiabatic processes"
FAN Seminar
21/05/2012

Afonin, Sergey
(Saint Petersburg State U.)
"Soft wall model with inverse exponential profile as a
model for the axial mesons"
HEP Seminar
22/05/2012

Das, Sumit R.
(University of Kentucky)
"Collective Fields in the Sp(N) Models and dS/CFT"
HEP Seminar
24/05/2012

El-Showk, Sheer (CEA Saclay)
"The Resurgent Bootstrap and the 3D Ising Model
(towards solutions of CFTs in D > 2)"
HEP Seminar
25/05/2012

Basilakos, Spyros
(Academy of Athens)
"The growth index of matter perturbations
and modified gravity"
HEP Seminar
29/05/2012

Papadimitriou, Ioannis
(IFT UAM/CSIC)
"A 'double cover' for the SO(6)xSO(2)
symmetric sector of N=8 gauged supergravity
in four dimensions"
HEP Seminar
31/05/2012

Pujolas, Oriol (UAB)
"Emergent Lorentz Invariance"
HEP Seminar
01/06/2012

Gabrielli, Emidio (NICPB Estonia)
"Fermiophobic Higgs scenarios at the LHC"
HEP Seminar
07/06/2012

Epelbaum, Evgeni
(Ruhr U., Bochum)
"A renormalizable EFT approach to NN scattering with
nonperturbative pions"
HEP Seminar
08/06/2012

Routray, T.R. (U. Sambalpur)
"Momentum and density dependence of nuclear mean
field and equation of state of nuclear matter"
FAN Seminar
20/06/2012

Bekaert, Xavier (LMPT Tours)
"Towards a bulk dual of the unitary Fermi gas"
HEP Seminar
21/06/2012

Holl, Berry (Lund Observatory, Lund, Sweden)
"Characterizing the astrometric errors in
the Gaia catalogue"
DAM Seminar
25/06/2012

Gomis, Jaume (Perimeter Institute)
"Exact Results in D=2 Supersymmetric Gauge Theories"
HEP Seminar
26/06/2012

Beane, Silas (U. New Hampshire)
"Nuclear physics from first principles: a status report"
FAN Seminar
26/06/2012

Dudas, Emilian (Ecole Polytechnique & Orsay, LPT)
"Universal gravitational contributions to scalar masses"
HEP Seminar
28/06/2012

Cantó, Jorge (Instituto de Astronomía, UNAM, México)
"Jets de estrellas jóvenes: Teoría"
DAM Seminar
03/07/2012

Miller-Jones, James (International Centre for Radio
Astronomy Research - Curtin University, Australia.)
"Astrometry of XRBs and what one can do with it"
DAM Seminar
04/07/2012

Comerón, Sebastien (Korea Astronomy and Space
Science Institute, University of Oulu)
"A deeper look on thick discs using data from the Spitzer
Survey of Stellar Structure in Galaxies (S4G)"
DAM Seminar
16/07/2012

Schat, Carlos (Ohio U. & Buenos Aires U.)
"Testing quark forces in baryons using the 1/Nc expansion"
HEP Seminar
27/09/2012

Batta Márquez, Aldo Alberto (Instituto de Astronomía,
Universidad Nacional Autónoma de México)
"Cooling induced structures in the collapsar model"
DAM Seminar
02/10/2012

Kniehl, Bernd (Hamburg U.)
"Heavy-quarkonium theory in the LHC era"
HEP Seminar
04/10/2012

Moran, James M.
(Harvard-Smithsonian Center for Astrophysics)
"Dinnertime for SgrA* (The Black Hole in the Center of
OUR Galaxy)"
DAM Seminar
05/10/2012

Hiyama, Emiko (RIKEN, Japan)
"Gaussian Expansion Method and application to 4He
tetramer system"
FAN Seminar
08/10/2012

Donos, Aristomenis (Imperial College, UK)
"Spatial modulation in AdS/CFT"
HEP Seminar
18/10/2012

G. Cámara, Pablo (ICCUB)
"Non-Abelian discrete gauge symmetries in String Theory"
HEP Seminar
19/10/2012

García, Miguel Ángel (ICCUB)
"Strongly correlated ultracold bosons as impurities immersed in a Bose-Einstein condensate"
FAN Seminar
24/10/2012

Hoyos, Carlos (Tel-Aviv U.)
"Chiral magnetic effect in holography"
HEP Seminar
25/10/2012

Russo, Jorge (ICCUB)
"Large N limit of super Yang-Mills theories from localization"
HEP Seminar
26/10/2012

Faedo, Anton (Swansea U.)
"RG flows as domain walls of N=4 supergravity"
HEP Seminar
08/11/2012

Carbone, Arianna (ICCUB)
"High momentum components in the nuclear symmetry energy"
FAN Seminar
08/11/2012

Planells, Xumeu (ICCUB)
"Study of local parity breaking in heavy ion collisions"
HEP Seminar
09/11/2012

Withers, Benjamin (Durham U.)
"Stars, Superfluids and Supergravity"
HEP Seminar
15/11/2012

Mateo, David (ICCUB)
"Desorption dynamics of photoexcited atoms in helium nanodroplets"
FAN Seminar
15/11/2012

Emparan, Roberto (ICCUB)
"Viscous fluid, elastic solid: a general framework for the effective dynamics of black objects"
HEP Seminar
16/11/2012

Chowdhury, Borun (U. of Amsterdam)
"Unitarity and fuzzball complementarity: Alice fuzzes but may not even know it!"
HEP Seminar
22/11/2012

Myers, Robert C. (Perimeter Institute)
"On the Architecture of Spacetime: Holography, Entanglement, c-theorems and Black Holes"
HEP Seminar
23/11/2012

Koenigsberger, Gloria
(Instituto de Ciencias Físicas, UNAM, México)
"HD5980: A prototype for consecutive LBV and supernova events"
DAM Seminar
26/11/2012

Monteiro, Ricardo (Copenhagen U.)
"Hairy black holes and solitons in global AdS_5"
HEP Seminar
29/11/2012

Tarrio, Javier (ICCUB)
"Describing holographically D3/D7 plasmas: successes and limitations"
HEP Seminar
30/11/2012

Obers, Niels
(Niels Bohr Inst., U. of Copenhagen)
"Blackfolds and thermal probe branes in string theory"
HEP Seminar
30/11/2012

Benicasa, Paolo
(U. de Santiago de Compostela)
"A holographic perspective on the high baryon density regime in field theories at strong coupling"
HEP Seminar
13/12/2012

At ICCUB

Figueras, F.; Luri, X.
Members of the organizing committee
International workshop
"Galaxy Modelling with a Gaia mock catalogue"
Faculty of Physics, UB
29/02/2012 - 02/03/2012
https://gaia.am.ub.es/Twiki/bin/view/WS2GaiaMock

Ribó, J.M.
Organizer
National meeting "AYA2009-13920-C02 Resultats I Prospectives; Trobada oberta"
Dep. de Química Orgànica, UB;
Dep. de Física Aplicada I Òtica, UB;
Centro de Astrobiología, CSIC-INTA
10/05/2012 - 11/05/2012
http://icc.ub.edu/documents/SeminariAYA2009.pdf

Aran, A.; Sanahuja, B.
Members of the organizing committee
International meeting "SPACECAST"
Dept. Astronomia i Meteorologia, UB
21/05/2012-24/05/2012

Sanahuja, B.; Àgueda, N.
Members of the organizing committee
International meeting "Meeting SEPserver"
Dept. Astronomia i Meteorologia, UB
17/09/2012-20/09/2012

Magas, V.; Parreño, A.; Polls; A. Ramos, A.
Members of the local organizing committee "HYP 2012: XI International Conference on Hypernuclear and Strange Particle Physics"
Cosmocaixa
01/10/2012 - 05/10/2012
http://icc.ub.edu/congress/HYP2012

Ramos, A.
President of the organizing committee
"HYP 2012: XI International Conference on Hypernuclear and Strange Particle Physics"
Cosmocaixa
01/10/2012 - 05/10/2012
http://icc.ub.edu/congress/HYP2012

Group picture of HYP 2012 at Cosmocaixa

Fernández, D.; Fröb, M.
Organizers
International Meeting "Barcelona Postgrad Encounters on Fundamental Physics"
Faculty of Physics, UB
17/10/2012 - 19/10/2012
http://ffn.ub.edu/bcn-encounters

Group picture of "Barcelona Postgrad Encounters on Fundamental Physics" at the Physics Faculty, UB

Paredes, J.M.; Bosch-Ramon, V.; Ribó, M.; Munar-Androver, P.; Marcote, B.; Paredes-Fortuny, X.; Frutos, A.
Members of the local organizing committee
International workshop "Exploring the Non-thermal Universe with Gamma Rays "
Faculty of Physics, UB
06/11/2012 - 09/11/2012
http://icc.ub.edu/congress/FAA60

Paredes, J.M.; Bosch-Ramon, V.
Members of the organizing committee
International workshop
"Exploring the Non-thermal Universe with Gamma Rays "
Faculty of Physics, UB
06/11/2012 - 09/11/2012
http://icc.ub.edu/congress/FAA60

Paredes, J.M.
President of the Scientific Committee
International workshop
"Exploring the Non-thermal Universe
with Gamma Rays "
Faculty of Physics, UB
06/11/2012 - 09/11/2012
http://icc.ub.edu/congress/FAA60

FAA60 session at the Physics Faculty, UB

Paredes, J.M.
Chair of the panel
"XMM-Newton B2 panel
meeting for the AO-12 review"
Dept. Astronomia i Meteorologia, UB
12/11/2012-14/11/2012

Mescia, F.; Espriu, D.
Organizers
National meeting
"12a Trobada de Nadal de Física Teòrica"
Faculty of Physics, UB
19/12/2012 - 20/12/2012
http://www.ecm.ub.es/~mescia/2012

Salvador, E.; Graciani, R.
Organizers
National conference "Iniciativa SATLANTIS"
Antoni Caparrós Auditorium, PCB
21/12/2012
http://icc.ub.edu/congress/SATLANTIS

At other institutions

Gómez, G.
Member of the organizing committee
International conference "1st IAA Conference on
Dynamics and Control of Spacecrafts"
Hotel HF Ipanema Porto (Portugal)
19/03/2012 - 21/03/2012
http://www.astrodynamics.org.pt/index.html

Jordi, C.
President of the organizing committee
National meeting
"La contribución de las ICTS españolas a la misión Gaia de
ESA. Reunión abierta de la RIA conjunta de La contribución
de las ICTS españolas a la misión Gaia de ESA
CDTI (Madrid, Spain)
21/03/2012 - 23/03/2012
https://gaia.am.ub.es/Twiki/pub/RecGaia/RiaICTS/RIA-
Gaia-Conclusiones.pdf

Figueras, F.
President of the organizing committee
National meeting
"La contribución de las ICTS españolas a la misión Gaia de
ESA. Reunión abierta de la RIA conjunta de La contribución
de las ICTS españolas a la misión Gaia de ESA"
CDTI (Madrid, Spain)
21/03/2012 - 23/03/2012
https://gaia.am.ub.es/Twiki/pub/RecGaia/RiaICTS/RIA-
Gaia-Conclusiones.pdf

Graugés, E.
Member of the organizing committee
"XL International Meeting on Fundamental Physics"
Centro de Ciencias de Benasque Pedro Pascual
(Huesca, Spain)
24/05/2012 - 03/06/2012
http://benasque.org/2012imfp/:

Jordi, C.
Member of the organizing committee
"The metallicity distribution in the Milky Way discs
(Bolonia, Italia) "
University of Bologna
29/05/2012 - 31/05/2012
http://www.bo.astro.it/great-esf-gradient/

Torra, J.
Member of the organizing committee
International meeting
"DPAC. CU3 meeting 2012"
ESA (Vilanova de Gaia, Portugal)
13/06/2012 -15/06/2012
http://www.rssd.esa.int/wikiSI/index.php?instance=Gai
a&title=CU3:Core_Processing:Meetings:CU3M7

Jordi, C.
Member of the organizing committee
"EWASS 2012: Gaia Research for
European Astronomy Training "

Pontificia Università Lateranense
(Roma, Italy)
04/07/2012 - 06/07/2012
http://www.ifsi-roma.inaf.it/ewass2012/

Jordi, C.
Member of the organizing committee
"X Reunión Científica de la Sociedad
Española de Astronomía"
University of Valencia (Spain)
09/07/2012 - 13/07/2012
http://www.sea-astronomia.es/drupal/SEA2012

Paredes, J.M.
President of the organizing committee
"5th International Symposium on High-Energy
Gamma-Ray Astronomy (Gamma2012)"
Max-Planck-Institut für Kernphysik
(Heidelberg, Germany)
09/07/2012 - 13/07/2012
http://www.mpi-hd.mpg.de/hd2012/
pages/news.php

Peñaranda, S.
International workshop
"Latinoamerican Workshop on High Energy Physics:
Particles and Strings"
La Habana, Cuba
15/07/2012 – 21/07/2012
http://dftuz.unizar.es/whepcuba2012/overview

Solà, J.
Co-organizator and member of the International Committee
IIRGAC 2012: III International Conference on Quantum Theories
and Renormalization Group in Gravity and Cosmology"
Paraty, Brasil
29/07/2012 - 03/08/2012
https://sites.google.com/site/3rdirgac/

Jordi, C.
Member of the organizing committee
"IAU Simposio 289 'Advancing the physics of cosmic
distances' en la XXVIII Asamblea General de la IAU"
IAU General Assembly, National Convention Centre
(Beijing, China)
27/08/2012 - 31/08/2012
http://www.mporzio.astro.it/IAUS289/www/Home.html

Luri, X.
Member of the organizing committee
International workshop
"Gaia Astro-Visualisation School"
Astronomy Department, University of Washington
(Seattle, USA)
14/09/2012 - 18/09/2012
http://great.ast.cam.ac.uk/Greatwiki/GreatItn/
VizSchoolSep2012

Public Outreach

Outreach Events

Talks

"El nostre lloc a l'espai"
Speaker: Carrasco, J.M.
Faculty of Physics, UB
Date: 10/01/12

"Xerrada sobre astronomia"
Speaker: Carrasco, J.M.
IES Jaume Salvador i Pedrol
(Sant Joan Despí)
Date: 30/01/12

"Pensant l'Univers"
Speaker: Carrasco, J.M.
IES Jaume Salvador i Pedrol (Sant Joan Despí)
Date: 30/01/12

"Com neixen, viuen i moren les estrelles"
Speaker: Jordi, C.
Club Muntanyenc Sant Cugat
Date: 20/02/12

"Estem sols a l'Univers?"
Speaker: Solanes, J.M.
Associació Dones d'Ara (La Garriga, Barcelona)
Date: 22/02/12

"Xerrada sobre relativitat"
Speaker: Luri, X.
Col·legi major Ramon-Llull (Barcelona)
Date: 06/03/12

"Passat, present i futur del nostre Sol"
Speaker: Jordi, C.
Aules d'extensió universitària (Sabadell)
Date: 12/03/12

"El trànsit de Venus"
Speaker: Carrasco, J.M.
INS Montmeló (Barcelona)
Date: 11/04/12

"Neutrins superlumínics i relativitat"
Speaker: Luri, X.
Date: 07/05/12

"Un viatge al centre de la Via Làctia"
Speaker: Jordi, C.
Associació Astronòmica Valldoreix-Sant Cugat
Date: 24/05/12

"La missió Gaia: Composició, estructura i evolució de la nostra Galàxia"
Speaker: Portell, J.
Universitat Catalana d'Estiu de la Natura (UCEN, Berga)
Date: 13/06/12

"Xerrada sobre astronomia"
Speaker: Luri, X.
Ajuntament de Barcelona, Festa de la ciència al parc de la Ciutadella
Date: 16/06/12

"Cosmogènesi: l'origen de l'univers"
Speaker: Solanes, J.M.
Pavelló de Suècia (Berga)
Date: 13/07/12

"Les distàncies còsmiques"
Speaker: Jordi, C.
Ajuntament de Sant Cugat
Date: 15/09/12

"Un gra de sorra a la platja còsmica"
Speaker: Luri, X.
Parc Astronòmic Montsec (Àger, Lleida)
Date: 20/10/12

"Gaia, observant a un milió i mig de quilòmetres"
Speaker: Torra, J.
Parc Astronòmic Montsec (Àger, Lleida)
Date: 21/10/12

"La missió Gaia"
Speaker: Carrasco, J.M.
Agrupació astronòmica ÀSTER
Date: 15/11/12

"La Galàxia en un petabyte"
Speaker: Portell, J.
Parc Astronòmic Montsec (Àger, Lleida)
Date: 17/11/12

"Gaia mira amb mil milions d'ulls"
Speaker: Jordi, C.
Parc Astronòmic Montsec (Àger, Lleida)
Date: 18/11/12

"El telescopi Fabra-ROA Montsec (TFRM): un instrument per a la cerca d'escombraries espacials, exoplanetes, NEOs, i fonts d'alta energía"
Speaker: Núñez, J.
Societat Catalana de Física,
Cicle Física Oberta 2012-2013
Date: 20/11/12

"Cada estrella és un món: de l'infantesa a la mort"
Speaker: Figueras, F.
Parc Astronòmic Montsec (Àger, Lleida)
Date: 01/12/12

"El Sistema Solar i altres planetes atrapats per estrelles"
Speaker: Masana, E.
Parc Astronòmic Montsec (Àger, Lleida)
Date: 02/12/12

"Visita d'un astrònom a les classes 'La Lluna' i 'Les estrelles' "
Speaker: Carrasco, J.M.
CEIP Baldiri i Reixac (Badalona)
Date: 13/12/12

"La Galàxia en un petabyte"
Speaker: Portell, J.
Agrupació Astronòmica d'Osona (Vic)
Date: 15/12/12

"El trànsit de Venus 2012 i la mida del Sistema Solar"
Speaker: Jordi, C.
Aules de la gent gran de la UB
Date: February 2012

"Naixement, vida i mort de les estrelles"
Speaker: Jordi, C.
Centre Cívic de Torre Llobeta (Barcelona)
Date: March 2012

"Un viatge al centre de la Galàxia"
Speaker: Jordi, C.
Centre Cívic de Torre Llobeta (Barcelona)
Date: October 2012

"The Sun and the Heliospheric Physics and Space Weather Group of the University of Barcelona"
Speaker: Rodríguez, R.
Participation in the project "Joves i Ciència" of Catalunya-Caixa
Date: 2012

Exhibitions

"Amb A d'AstrònomA"
ICCUB Organizers: Olarte, B.; Figueras, F.; Jordi, C.; Balaguer, M.D.

Places:

- Centre Dolors Piera d'Igualtat d'Oportunitats i Promoció de les Dones, (Lleida)
1-30 March 2012

- Casal Font d'en Fargues (Barcelona)
18/04/2013-15/05/2013

- Casa de la Dona (Quart de Poblet, València)
November 2012

"Amb A d'Astrònoma" at Casa de la Dona, Quart de Poblet

"Les distàncies còsmiques"
ICCUB Organizers: Olarte, B.; Figueras, F.; Jordi, C.; Balaguer, M.D.
Places:

- Parc Astrònomic Montsec (Àger, Lleida)
29/08/2012-31/08/2012

- Ajuntament de Sant Cugat (Sant Cugat, Barcelona)
15/09/2012-30/09/2012

- Institut Lluís Vives (Barcelona)
16/11/2012-25/11/2012

"Les distàncies còsmiques" at Institut Lluís Vives

"De la Terra a l'Univers"
ICCUB Organizers: Olarte, B.; Figueras, F.; Jordi, C.; Balaguer, M.D.

Places:

- Museu Palau Mercader (Cornellà de Llobregat)
30/09/2012-25/11/2012

"De la Terra a l'Univers" at Museu Palau Mercader

Workshops and Schools

"Taller de Física de Partícules"
Masterclass for High School Students
Date: 7 and 13 March 2012
Organization: ICCUB
ICCUB Participants: Garrido, Ll., Graciani, R.; Graugés, E.; Rives, V.; Ruiz, H. Vázquez, R.

"Physis 2012', Quart campus d'estiu de Física per a estudiants de Batxillerat"
Summer Course
Date: 18/06/12-22/06/12
Place: Facultat de Física
Organization: Physics Faculty, University of Barcelona
ICCUB Participants: Solanes, J.M.; Carrasco, J.M.

"Experiments de Física"
Laboratory sessions
Date: 23/01/12-03/02/12
Place: Facultat de Física
Organization: Physics Faculty, University of Barcelona
ICCUB Participants: Carrasco, J.M.; Viñas, J.; Udina, M.; Fortuny, D.; Moldón, J.; Roca, S.

"Un cop d'ull a l'Univers"
Summer Course
Date: 08/07/12-10/07/12
Place: Centre d'Observació de l'Univers, Àger
Organization: University of Lleida
ICCUB Participants: Carrasco, J.M.

Astronomy Sessions

"Observació del Sol",
Speaker: Carrasco, J.M.
Date: 2012/02/03
Place: Facultat de Física, UB (Barcelona)

"Nit d'astronomia a les caves Nadal"
Speaker: Carrasco, J.M.; Vilella, G.
Dates: 2012/03/03
Place: Caves Nadal surroundings

"Observació popular del trànsit de Venus per serviastro.am.ub.es des del castell de Montjuic de Barcelona"
Speaker: Carrasco, J.M.
Dates: 06/06/2012
Place: Castell de Montjuic (Barcelona)

"Nit d'astronomia i estels al parc natural de Montserrat"
Speaker: Romero, M.; Monguió, M.
Dates: 2012/07/21
Place: Parc de Montserrat (Collbató)

"Observació popular"
Speaker: Carrasco, J.M.
Dates: 2012/07/27
Place: La Pobla de Cèrvoles

"Observació del Sol en el marc de la II Festa dels Amics de l'Hospital Sant Joan de Déu"
Speaker: Monguió, M.; Masana, E.
Dates: 22/09/12
Place: Palau Robert (Barcelona)

Other events

Venus transit online retransmission from Svalbard Islands (Norway)
Date: 5-6/06/2012:
ICCUB Organizers: Balaguer, L.; Jordi, C.; Masana, E.

The Venus transient retransmission at Serviastro Site

Printed and Electronic Media

Press releases

"S'identifica el moment d'explosió de material en un forat negre"
Date: 11/01/12
Authors: Bonmatí, B.; Migliari, S.
http://www.ub.edu/web/ub/ca/menu_eines/noticies/2012/01/014.html

"L'observació de l'Univers permet precisar el límit de la massa dels neutrins"
Date: 12/01/2012

Authors: Bonmatí, B.; De Putter, R.
http://www.ub.edu/web/ub/ca/menu_eines/noticies/2012/01/018.html

"Article a 'Nature' sobre l'origen de les supernoves de tipus Ia"
Date: 13/01/2012
Authors: Bonmatí, B.; Ruiz-Lapuente, P.
http://www.ub.edu/web/ub/ca/menu_eines/noticies/2012/01/023.html

"Es posa en marxa el projecte Spacecast per pronosticar el temps espacial"
Date: 12/03/2012
Authors: Bonmatí, B.; Àgueda, N.; Aran, À.; Sanahuja, B.
http://www.ub.edu/web/ub/ca/menu_eines/noticies/2012/03/032.html

"Escolars d'arreu de Catalunya preparen el proper trànsit de Venus amb activitats elaborades per la UB"
Date: 22/05/2012
Authors: Bonmatí, B.; Balaguer, D.; Jordi, C.; Massana, E.; Olarte, B.
http://www.ub.edu/web/ub/ca/menu_eines/noticies/2012/05/078.html

"La Universitat de Barcelona i el Parc Astronòmic del Montsec retransmetran des de Noruega, en directe, el darrer trànsit de Venus d'aquest segle"
Date: 05/06/2012
Authors: Bonmatí, B.; Balaguer, D.; Jordi, C.; Massana, E.; Olarte, B.
http://www.ub.edu/web/ub/ca/menu_eines/noticies/2012/06/001.html

Crònica del trànsit de Venus des de Noruega
Date: 06/06/2012
Authors: Bonmatí, B.; Balaguer, D.; Jordi, C.; Massana, E.; Olarte, B.
http://www.ub.edu/web/ub/ca/menu_eines/noticies/2012/06/018.html

"UB's lecturer Licia Verde, one of the recipients of the 2012 Gruber Cosmology Prize"
Date: 22/06/2012
Authors: Bonmatí, B.; Salvador, E.
http://www.ub.edu/web/ub/en/menu_eines/noticies/2012/06/071.html

"Descobreixen la causa del fenomen estel·lar més brillant de la història, la supernova de l'any 1006"
Date: 27/09/12
Authors: Bonmatí, B.; Ruiz-Lapuente, P.
http://www.ub.edu/web/ub/en/menu_eines/noticies/2012/09/049.html

"L'ESA presenta la missió Gaia, en la qual participen 30 investigadors de la UB"
Date: 23/10/12
Authors: Bonmatí, B.; GAIA group
http://www.ub.edu/web/ub/ca/menu_eines/noticies/2012/10/096.html

"Presentació de la participació espanyola en la missió Gaia"
Date: 30/10/12
Authors: Bonmatí, B.; GAIA group
http://www.ub.edu/web/ub/ca/menu_eines/noticies/2012/Fotonoticies/10/018.html

"An international team of astronomers measures the Universe's deceleration before dark energy took over"
Date: 13/11/2012
Authors: Bonmatí, B.; Miralda, J.; SLOAN Collaboration
http://www.ub.edu/web/ub/en/menu_eines/noticies/2012/11/034.html

Publications in press

From press release ***"S'identifica el moment d'explosió de material en un forat negre":***

11/01/2012, Sinc (Electronic press), "Identificado el momento en que explosiona material en un agujero negro".

11/01/2012, La Vanguardia.com (Electronic press), "Identificado el momento de la explosión de material de un agujero negro".

11/01/2012, El Mercurio Digital (Electronic press), "Identifican el momento en que explota material en un agujero negro".

11/01/2012, El Imparcial (Electronic press), "Identificado el momento de la explosión de material de un agujero negro".

13/01/2012, Universia (Electronic press), "Se identificó el momento en que explosiona material en un agujero negro".

From press release ***"L'observació de l'Univers permet precisar el límit de la massa dels neutrins":***

13/01/2012, El Punt Avui (Press), "Experts del CSIC calculen la massa dels neutrins".

12/01/2012, SINC (Electronic press), "Los tres neutrinos juntos tienen una masa dos millones de veces menor que el electrón".

12/01/2012, La Vanguardia.com (Electronic press), "Investigadores obtienen el límite más preciso hasta la fecha de la masade los neutrinos observando el Universo".

From press release ***"Es posa en marxa el projecte Spacecast per pronosticar el temps espacial":***

12/03/12, elPeriódico.cat (Electronic press), "Es posa en marxa el projecte Spacecast per al pronòstic del temps espacial".

12/03/2012, La vanguardia.com (Electronic press), "La peor tormenta solar de la historia causaría ahora 22.860 millones de pérdidas".

12/03/2012, La vanguardia.com (Electronic press), "La UB colabora en una investigación europea para pronosticar la meteorología espacial".

12/03/2012, elEconomista.es (Electronic press), "Si se repitiera la peor tormenta solar de la historia generaría pérdidas de 22.860 millones".

12/03/2012, elPeriódico.com (Electronic press), "La mayor tormenta solar registrada causaría ahora 23.000 millones de euros de pérdidas".

14/03/2012, Atlántico (Electronic press), "La peor tormenta solar de la historia causaría ahora 22.860 millones pérdidas".

16/03/2012, Amazings, Noticias de la Ciencia y la Tecnología (Electronic press), "Se pone en marcha el proyecto Spacecast para pronosticar el tiempo espacial".

From press release ***"Escolars d'arreu de Catalunya preparen el proper trànsit de Venus amb activitats elaborades per la UB":***

24/05/2012, Europa Press (Electronic press), "La UB desarrolla una web para ver en directo el último tránsito de Venus de este siglo".

29/05/2012, Ice People (Press, Norway), "Spotting Venus".

02/06/2012, El Mundo (Edicion nacional) (Press), "El 'paseo' de venus delante del Sol".

02/06/2012, La Mañana Diario de Ponent (Press), "El COU d'Àger retransmitirá el paso de Venus ante el Sol".

02/06/2012, La Mañana Diario de Ponent (Press), "La Picota".

02/06/2012, Segre (Ed. català) (Press), "Visió del trànsit de Venus des 'Montsec' ".

03/06/2012, La Tercera (Electronic press), "Guía para ver dos de los eventos astronómicos más importantes del año".

03/06/2012, Ara (Press), "El trànsit de Venus fa llum sobre misteris de l'univers".

04/06/2012, Teinteresa.es (Electronic press), "Internet permite a España ver el final del tránsito de Venus".

04/06/2012, El Punt Avui (Press), "Si voleu veure Venus transitar pel Sol, dimecres o mai més". Advised by: Carrasco, J.M.

From press release ***"La Universitat de Barcelona i el Parc Astronòmic del Montsec retransmetran des de Noruega, en directe, el darrer trànsit de Venus d'aquest segle":***

05/06/2012, Lainformación.com (Electronic press), "Venus pasará este martes por delante del Sol visto desde la Tierra".

05/06/2012, RTVE.es (Electronic press), "El último tránsito de Venus de este siglo se verá la madrugada de este martes desde la Tierra".

05/06/2012, Informativostelecinco.com (Electronic press), "Venus pasará hoy por delante del Sol visto desde la Tierra".

05/06/2012, Expansion.com (Electronic press), "La oportunidad del siglo para ver a Venus pasar por delante del Sol".

05/06/2012, Deia.com (Electronic press), "Venus pasará este martes por delante del Sol visto desde la Tierra".

05/06/2012, Notícias de Navarra (Electronic press), "Venus pasará este martes por delante del Sol visto desde la Tierra".

05/06/2012, Diario de Navarra (Press), "Venus pasea esta moche delante del Sol".

05/06/2012, El Periodico de Aragon (Press), "Una peca en el Sol".

05/06/2012, El Periodico de Catalunya (Ed. Catala) (Press), "Una piga al Sol".

05/06/2012, Ultima Hora (Press), "El paseo de Venus". From press release "Crònica del trànsit de Venus des de Noruega"

06/06/2012, Lainformación.com (Electronic press), "16.000 personas siguen en directo el tránsito de Venus en la web de la UB".

06/06/2012, Publico.es (Electronic press), "Venus pasea por delante del Sol".

06/06/2012, Diario De Pontevedra (Press), " El tránsito de Venus, una oportunidad única para la ciencia y la curiosidad humana".

07/06/2012, Bon Dia (Lleida) (Press), "El Parc del Montsec acosta el trànsit de Venus al món".

07/06/2012, El Punt Avui (Press), "Un Sol pigat massa efímer".

07/06/2012, La Mañana Diario de Ponent (Press), "Astrónomos de Montsec y Barcelona siguen a Venus desde Noruega".

07/06/2012, Segre (Ed. Catala) (Press), "Pendents d'un puntet".

From press release ***"Descobreixen la causa del fenomen estel·lar més brillant de la història, la supernova de l'any 1006":***

26/09/2012, El Pais.com (Electronic press), "La supernova del año 1006 se debió a la fusión de dos estrellas enanas blancas".

26/09/2012, El periodico.com (Electronic press), "Descubierto el origen de la inmensa supernova del año 1006, visible en toda la Tierra durante tres años".

26/09/2012, El Progreso (Galicia) (Electronic press), "La fusión de 2 enanas blancas causó la supernova más brillante conocida".

26/09/2012, Sinc (Electronic press), "La explosión estelar más brillante de la historia ya tiene explicación".

26/09/2012, La tercera (Electronic press), "Científicos aseguran que la fusión de dos enanas blancas causó la supernova más brillante jamás conocida".

26/09/2012, Física Hoy (Electronic press), "La explosión estelar más brillante de la historia ya tiene explicación".

27/09/2012, El Pais (Edición Nacional) (Press), "Choque de estrellas milenario".

27/09/2012, El Periodico de Catalunya (Press), "L'explosió estel·lar més gran".

27/09/2012, El correo gallego (Press), "La fusión de dos enanas blancas originó la supernova de 1006, el evento estelar más brillante conocido".

27/09/2012, Mediterráno (Castellón) (Press), "La fusión de dos enanas blancas causó la supernova más brillante".

27/09/2012, Madrid + D (Electronic press), "La fusión de dos enanas blancas causó la supernova más brillante de la historia".

27/09/2012, (Electronic press), "La fusión de 2 enanas blancas causó la supernova más brillante conocida".

29/09/2012, La vanguardia.com (Electronic press), "La fusión de dos enanas blancas causó la supernova más brillante conocida".

From press release ***"An international team of astronomers measures the Universe's deceleration before dark energy took over":***

BBC Mundo (Electronic press), "Un mapa en 3D para explorar la historia del Universo".

Audiovisual Media

"La misión Gaia: participación española"
Video-documental about the Gaia satellite.
Script, interviews and 3D animations by: Luri, X.; Czekaj, M.; Carrasco, J.M.; Jordi, C.; Figueras, F.; Torra, J.

Cartoon serial by DeBoom Studio
Colaboration as scientific adviser
Luri, X.

Radio and Television

"Calendaris"
With the participation of Luri, X.
Program: InfoNit (Barcelona TV)
Date: 29/02/2012

"Tempesta solar del 24 de gener de 2012"
With the participation of Sanahuja, B.
Date: 2012/01/26

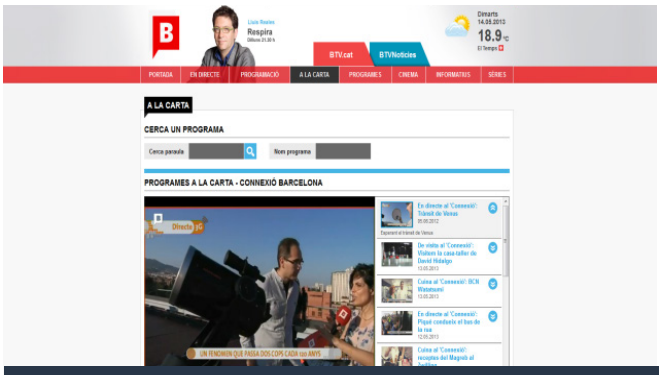
"L'activitat solar durant el 2012"
With the participation of Sanahuja, B.
Date: 21/01/2012

"Activitat solar i temps espacial"
With the participation of Sanahuja, B.
Program: Notícies de les 10 (BTV)
Date: 23/01/2012 http://www.btv.cat/alacarta/informatius/13470

"L'activitat solar el 7 de març de 2012"
With the participation of Sanahuja, B.
Program: El matí de Catalunya Ràdio (Catalunya Radio)
Date: 07/03/2012
http://www.catradio.cat/audio/617867/El-Sol-estaentrant-en-un-periode-dactivitat-que-pot-tenir-repercussions-en-el-nostre-entorn

"El trànsit de Venus"
With the participation of Carrasco, J.M.,
Program: Extraradi (COM Radio)
Date: 05/06/2012 http://www.am.ub.edu/twiki/pub/ServiAstro/MitjansInformacio050612/extraradi_050612_JMC_TransitVenus.mp3

"El trànsit de Venus"
With the participation of Carrasco, J.M
Program: Connexió Barcelona (BTV)
http://www.btv.cat/alacarta/connexio-barcelona/19002/



Josep Manel Carrasco at "Connexió Barcelona", BTV

"El trànsit de Venus"
With the participation of Carrasco, J.M.
Program: La Sexta - Notícias
Date: 05/06/201

"El trànsit de Venus"
With the participation of Carrasco, J.M.
Program: 8 al dia amb Josep Cuní (BTV)
Date: 06/06/2012 http://www.8tv.cat/8aldia/videos/per-a-que-serveix-lobservacio-del-transit-de-venus-entre-el-sol-i-la-terra

Josep Manel Carrasco at "8 al Dia", BTV

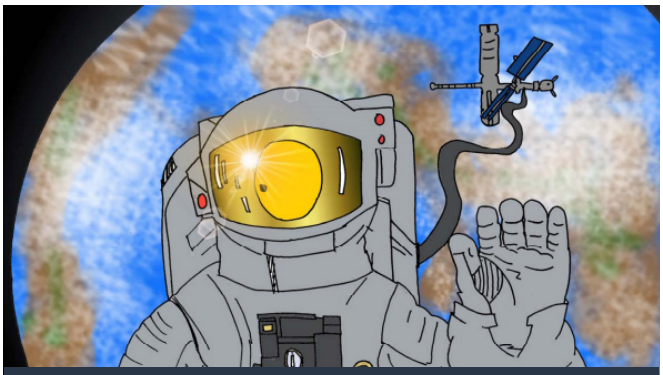
Web Pages

Serviastro Site
http://serviastro.am.ub.edu
Authors: Balaguer, L.; Jordi, C.; Masana, E.; Olarte, B.

Descobrint LHC
http://www.lhc.cat/
Authors: Frutos, A.; Ruiz, H

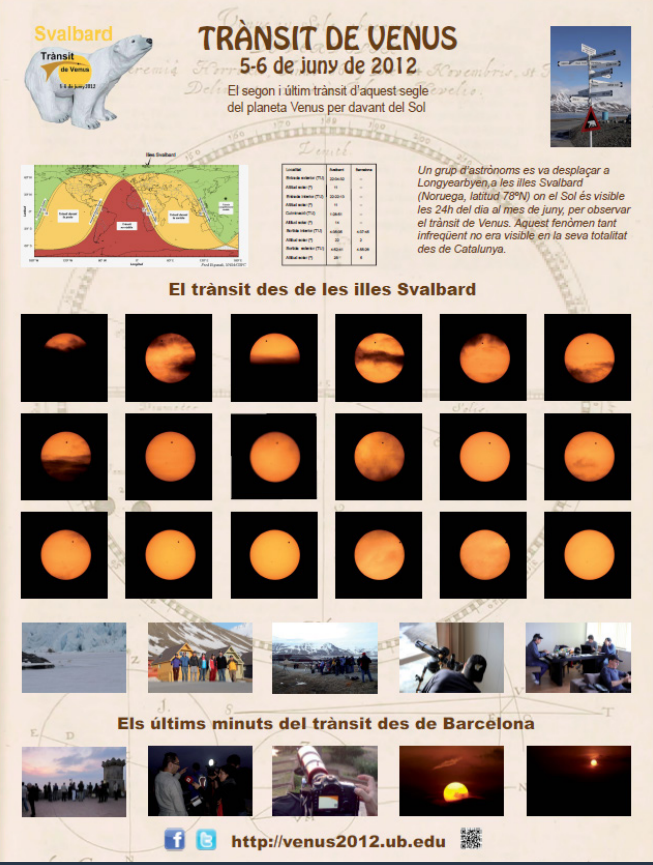
Pedagogical Material

"La Historieta del Trànsit de Venus"
Online publication
Author: Carrasco, J.M.
http://serviastro.am.ub.edu/twiki/pub/ServiAstro/TrVe/Historieta_cat3_A4.pdf.



La historieta del Trànsit de Venus

"Trànsit de Venus 2012"
Poster
Author: Olarte, B.



Poster of the transit of Venus

"La distància Terra-Sol"
Masana, E.
http://www.am.ub.edu/twiki/bin/view/ServiAstro/DistanciaTerraSol050612

"La mida del Sol "
Carrasco, J.M.
http://www.am.ub.edu/twiki/bin/view/ServiAstro/MidaSol050612

"La mida del Sistema Solar"
Masana, E.
http://www.am.ub.edu/twiki/bin/view/ServiAstro/MidaSistemaSol050612

"La distància a una estrella "
Jordi, C.
http://www.am.ub.edu/twiki/bin/view/ServiAstro/DistanciaEstrella050612

Other Public Outreach Activities

"Comprender la Física que nos rodea."
Solanes, J.M.
Programm "Campus Científicos de Verano" for high school students, by FECYT and ME

"Posta en marxa del FaceBook de la Sociedad Española de Astronomia "
Luri, X.

FUNDING

ICCUB Budget

Type of Income	Amount
Contract Program UB 2011-2012	68.071,04 €
Manager joint financing	22.970,26 €
IEEC Overheads	18.101,94 €
TOTAL	**109.143,24 €**

Group Project Funding

Type of Income	Amount
National Projects	1.860.166 €
European Projects	1.074.959 €
Complementary Actions	8.220 €
Contracts	51.680 €
Consolidated Groups	45.321 €
Other Grants	51.906 €
TOTAL	**3.092.252 €**

ICCUB Expenses

Type of Income	Contract Program UB	IEEC	Total
Personnel	74.204 €	14.641 €	88.845 €
Maintenance / Infrastructures / Inventory	786 €	211 €	997 €
Expendable	2.842 €	1.978 €	4.820 €
External Colloquia	2.673 €	1.148 €	3.821 €
Workshop Aids	403 €	0 €	403 €
Travels	271 €	0 €	271 €
Outreach	0 €	124 €	124 €
Other	9.862 €	0 €	10.814 €
TOTAL	**91.041 €**	**18.102 €**	**109.143 €**